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Business Process Automation in SMEs: A Systematic Literature Review

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ABSTRACT Business Process Automation has been gaining increasing importance in the management of companies and organizations since it reduces the time needed to carry out routine tasks, freeing employees for other, more creative and exciting things. It can be applied in the most varied business areas. Organizations from any sector of activity can also adopt it. Given these benefits, the granted success in transforming business processes would be expected. However, automation initiatives still fail. Adopting this technology can raise social, technological, ethical, methodical, and organizational issues. These facts have triggered the necessity for a summary that could extract more information about how to implement Business Process Automation (BPA), attending to the administrative processes, especially when applied in Small and Medium-sized Enterprises (SMEs). This study aims not only to review the available literature on how to implement BPA but also to typify the processes that can be automated, which technologies or tools exist for making that change, and influence factors in the procedure of BPA. We have covered more than 300 research papers published between 2016 and 2023 in reputable scientific data sources like Scopus, Web of Science/Clarivate, and ScienceDirect/Elsevier. The review revealed some paths for BPA, with some common steps. In addition, some common process characteristics fundamental to automation are exposed, as well as factors that are critical to an organization for successful automation. The results indicate that BPA is an established area in Business Process Management related to technologies/tools like Robotic Process Automation (RPA) or Cognitive-Robotic Process Automation (C-RPA), Workflow Management Systems (WfMS), Enterprise Resource Planning (ERP), and Blockchain. As far as we observed, this Systematic Literature Review (SLR) is a unique study that covers all the environmental variables for applying automation in business processes.

INDEX TERMS Digital Transformation (DX), Business Process Automation (BPA), Business Process Management (BPM), Model for BPA, Roadmap for BPA, Guidelines for BPA, Influencing factors in BPA, Robotic Process Automation (RPA), Cognitive-Robotic Process Automation (C-RPA), BPA in SMEs

I. INTRODUCTION

Digital Transformation (DX) is an area of interest for organizations of all sizes and sectors [1], and its growth is largely due to the global pandemic context (COVID-19) in the present past. Companies and organizations reinvented their work paradigm to continue providing services and products to markets [2]. DX can alter the value-creation process of the organization [1]. Morgan [3], in Forbes, mentioned that around 70% of companies had a DX strategy or were already implementing it, 21% thought they had completed their DX scheme, and there would be an investment of around $2 trillion in this transformation in 2019.

Paschek et al. [4] mentioned that business processes are combined with IT, which has become an important aspect of this study area. There are about 23.1 million Small and Medium-sized Enterprises (SMEs) (employees under 250) in the European Union [5]. In a study conducted among SMEs in the European Union in 2020, 43% have already adopted cloud computing [5]. According to Statista [5], with 22.6 million SMEs in the European Union, 34% have already adopted digital technology, 24% recognize that they need a digital tool, 10% consider adopting advanced digital technology, and 8% have the same perception as the previous group. However, due to a lack of digital literacy or funding, they do not know which one will be the most appropriate [5].

Administrative business process digitalization can involve automation or simple, more careful management. Business processes (administrative) can be defined as a collection of related activities (events), resources, and decisions that involve roles (actors) that lead to a valuable output, fulfilling customer needs and implementing the strategic objectives of the organization [4, 6-14]. Business processes are the arterial system in organizations and inter-organizational structures (essential for any competitive advantage) [4, 15-16]. Different performance dimensions characterize each process; for Denner et al. [8], it can be done with costs, flexibility, quality, and time.

Business Process Management (BPM) is essential for the constant improvement and study of the efficiency and effectiveness of an organization [8, 15, 17-23]. BPM must be used by SMEs and large enterprises [20], as their use is increasing [24] and adding value to the business process [25].

There are a lot of considerations about what BPM is: a multidisciplinary area that includes business process management, modeling, automation, execution, control, measurement, and optimization [26]; a set of components like modeling, analysis, deployment, execution, and monitoring [10]; the life cycle is a circular sequence of planning, analysis, design and modeling, implementation, monitoring and control, and refining the process [22]; is the use of concepts, methods, and techniques to plan, administrate, configure, and analyze the business process so that organizations can improve, manage, and control their business processes [8-9, 25, 27-28]; an involving element like external environment, organization ownership, strategic alignment, governance, methods, information technology, people, and culture [17]; a six-core element as strategic alignment, governance, methods, information technology, people, and culture [29].

To a successful BPM project, Gabryelczyk and Roztocki [30] described critical success factors like strategic alignment, project and change management, governance, performance management, methods, IT, people, culture, and communication. BPM plays an important role in the digital lifecycle of business processes that involve multiple participants and software systems and is considered a crucial enabler to increase the degree of business process automation to maintain competitiveness [31].

Ahmad and Van Looy [20] support the idea that the core of BPM (*automation and transformation of business processes*), with the emerging opportunities based on Information Technology (IT), can enhance business performance. BPM is related to Business Process Reengineering/Improvement (BPR/I) and Business Process Automation (BPA) [6, 9, 15].

BPR could improve the efficiency and effectiveness of the business process [32] and is not only about automation of the existing process but about complete redesign and change [33]. BPR must respect the structure of the organization, its resources (financial, human, and quality management experiences), its culture (risk attitude, employee, management support, innovation, strategic and business planning), and technology (IT/IS infrastructure, end-users skills, investment) [33].

Gartner [34] mentioned that (given the pandemic context), the use of process automation technologies would be an essential topic (90% of large companies worldwide would adopt this technology by 2022), with increasing importance for companies and the economy in general. These process automation technologies will automate critical business processes, freeing employees from the manual effort for other tasks [15, 34].

The technologies/tools that can be helpful to BPM and BPA, changing and improving business processes, are Blockchain [8, 10, 15, 19, 20, 35-36], Robotic Process Automation (RPA) [13, 15, 19, 21, 23, 37-48], Artificial Intelligence (AI) [13, 15, 19, 21, 23, 31, 36, 41-42, 44, 49-50], ERPs [21, 51-54], and Workflow Management/ Business Process Management Systems [12, 27, 36, 40-42, 51, 55-56]. It is usual to find a use of RPA technology in BPM and BPA [14-15, 23, 26, 31, 38, 48]. They complement each other [14, 31, 57]. *RPA complements BPM in many areas and vice versa but does not replace it* [14]. While BPM provides companies with actual knowledge about their business processes and workflows, RPA is concerned with developing bots whose function is to imitate human interaction with information systems [31].

Given the general benefits of adopting business process automation technologies/tools, automation can fail for various reasons. Stands out in the assessment of the process of digitalization, its documentation, its maturity, and, in short, the assessment of digital readiness; there is also the doubt in choosing the ideal process to automate; lack of know-how about the automation technology in the organization; definition of metrics for measuring the benefits achieved with automation, the definition of critical factors for success in automating and their implications, and lack of knowhow about the implementation methodology [58].

In this present work, we will find the reasons for automation failure. Combining the factors that contribute to the failure in the procedure of process automation, a roadmap that guides this entire process, from the as-is study, the to-be study, adoption, and follow-up, is necessary. This work will be guided by the research goal: *What methodological support can be given to SMEs for successfully adopting process automation tools?*

A. RELATED SURVEYS

As a result of the search, the authors of this study considered that a description of the protocol of literature review or systematic literature review (SLR) in 17 publications – [15, 31, 39, 42, 47-48, 59-69], related to the area of study in this SLR, Business Process Automation in SMEs (Table I). The other studies have a theoretical background based on literature only for contextualizing the core of each paper. This section presents a summary of each of them for this SLR.

TABLE I
SLR IN THE AREA OF BUSINESS PROCESS AUTOMATION FOUNDED IN THIS WORK

Authors	Process for automation RQ1	Technology/tool RQ2	Roadblocks RQ3	Benefits RQ4	How to? RQ5
Stravinskienė and Serafinas [15]	x	RPA	x	x	
Wewerka and Reichert [31]	x	RPA	x	x	x
Noppen et al. [39]	x	RPA	x	x	x
Engel et al. [42]		Cognitive Automation	x		
Herm et al. [47]	x	RPA	x		x
Enríquez et al. [48]	x	RPA		x	x
Vanwersch et al. [59]			x	x	
Moreira et al. [60]	x	RPA	x	x	
Costa et al. [61]		RPA	x	x	x
Syed et al. [62]	x	RPA/IPA	x	x	
Ng et al. [63]	x	IPA	x	x	
Sobczak [64]		RPA	x	x	
Hindel et al. [65]	x	RPA	x	x	
Kouriati et al. [66]		ERP	x	x	
Zayas-Cabán et al. [67]	x	Workflow automation	x	x	
Plattfaut et al. [68]	x	RPA	x	x	
Santos et al. [69]	x	RPA		x	

Some SLRs (Table I) are based on RPA [15, 31, 39, 47-48, 60-62, 64-65, 68-69], others in IPA [42, 62, 63], and others in workflow automation [67] or ERP [66], but not on automation technology in general. Relatively to the factors that can influence automation positively, the focus of the CSFs listed in [66] is related to ERP Systems, and in [68], it is related to RPA. So far, we have not found any SLR in the field of business process automation in general that lists the characteristics of the process candidate to automate, regardless of the type of technology; that identifies and characterizes the automation technologies that SMEs can adopt; that lists the main barriers, risks, and limitations in business process automation in SMEs; that characterize the evaluation of the success of the business process automation in SMEs; that identify and compare the existent framework, guidelines, methodologies or roadmaps to apply business process automation in SMEs, detecting their gaps.

With this SLR, the authors aim to document the overall process and knowledge about business process automation. In addition to the general enrichment of knowledge in business process automation research, this SLR also aims to mitigate the lack of studies in the area involving SMEs.

B. CONTRIBUTION

With this SLR paper, we intend to compact the results of our search on Business Process Automation in SMEs.

This article validates some gaps in the area and surveys the state of the art in business process automation. So far, we have not found any SLR in the field of business process automation that gathers information on aspects that are decisive for the success of the procedure, how to prepare and how to evaluate the post-procedure, and how to do it, with a focus on automation technologies/tools, and not just one. Some SLRs are based on RPA but not on automation technology/tools in general. This SLR pretends to give answers to these doubts: what process automates, which type of technology can the organization use, what factors must be aware of to mitigate the failure of the procedure, how can the SME evaluate the automation impact, and which steps the organization automates must do for success in automation.

Our study focuses on the application of automation to SMEs. However, this SLR found that this kind of study in SMEs is scarce. Therefore, publications deemed interesting were included in this study, regardless of the business area and type of organization.

The article is structured as follows: Section II presents the research methodology used to fulfil the purpose mentioned above; Section III describes and presents the findings for each of the five research questions. Section IV is the conclusion, where the main lines in this review are mentioned. This paper ends with announcing the future work in this path by the authors and some highlights for future investigations.

II. RESEARCH METHODOLOGY

We have selected the methodology Systematic Literature Review (SLR) or Systematic Review (SR) to achieve the identification, evaluation, and interpretation of all available research relevant to a research question, topic area, or phenomenon of interest [70-71]. The presented SLR was performed with Kitchenham and Charters [71] guidelines for Systematic Literature Reviews. These guidelines materialized all the procedures in three parts: planning, conducting, and reporting (Figure 1). Each part lists tasks to perform.

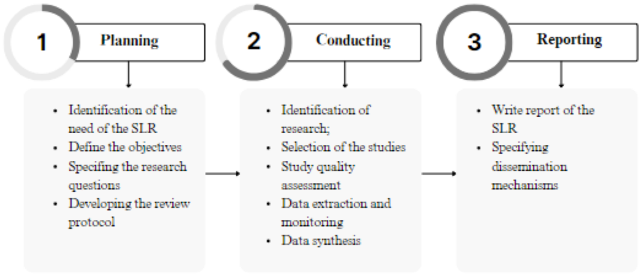

FIGURE 1. **Phases and tasks for an SLR (adapted from [60]).**

In the first phase, planning, it is necessary to identify the need for a review, define the goals/objectives, determine the research questions, and develop the review protocol. This protocol includes defining the search strategy, exclusion and inclusion criteria, and the extraction of data strategy. In the intermediate step, conducting, the review protocol is applied to filter articles according to the inclusion and exclusion criteria to obtain important information regarding the goal of the SLR. Then, the quality assessment is applied to exclude more literature, and the data extraction and synthesis happens. Reporting is the last phase of SLR by Kitchenham [70]. This is where we can find all the key findings extracted from the literature. Between January and April 2023, several searches were conducted in digital reference libraries such as Scopus, Web of Science/Clarivate, and Science Direct/Elsevier to find the literature in this SLR.

A. RESEARCH QUESTIONS

The aim of this SLR goes beyond providing a general overview of the current state of the art in the area of Business Process Automation in SMEs. The research goal mentioned in the Introduction Section is "*What methodological support can be given to SMEs for successfully adopting process automation tools?*". Given the inherent complexity of this question, the authors conceptually divided the issue into five research questions, as shown in Table II.

This SLR intends to provide answers regarding how to choose the processes to automate, how to choose the correct technology/tool for automation, how to implement a procedure of automation in business processes, and what challenges SMEs must be concerned about in terms of business process automation. Probably the most important topic related to how to implement automation in SMEs. As a result, this research plans to collect data that can answer the research questions, as in Table II.

TABLE II
RESEARCH QUESTIONS AND THEIR MOTIVATIONS

Research Question (RQ)	Description and motivation
(RQ1) How to typify the processes for automation?	Identify the characteristics most cited that a business process must have to be a candidate for automation, regardless of the type of automation technology.
(RQ2) What are the main technologies/tools to support business process automation?	Identify and characterize the automation technologies/tools that organizations can adopt.
(RQ3) What are the main barriers, risks, and limitations in business process automation in SMEs?	Organizations must pay attention to some items and areas to achieve automation procedures successfully. This search intent resumes them.
(RQ4) How to evaluate the success of business process automation in SMEs?	After automation, the organization must check whether there have been improvements in its indicators compared to the scenario characterized before automation. This RQ aims to determine indicators for this analysis.
(RQ5) What frameworks exist to support process automation for SMEs?	To guide organizations in their automation procedure, we intend to identify and compare existing frameworks, guidelines, or roadmaps, detecting gaps.

B. RESEARCH STRATEGY

Developing the review protocol, as suggested in Kitchenham and Charters [71], we have defined the PICOC (Population, Intervention, Comparison, Outcome, and Context) items as defined like:

- Population: SMEs.
- Intervention: Framework for Business Process Automation in SMEs.
- Comparison: Frameworks, methodologies, roadmaps, or guidelines for Business Process Automation.
- Outcome: Positive and negative key aspects of automation; challenges for automation; technology/tools for automation; comparison of the frameworks for automation that already exist.
- Context: Business Process Management and Business Process Automation.

The data sources and search characteristics are common for all the research questions. We can consult these two elements in the next table (Table III). We have selected Scopus, Web of Science (WoS), and Science Direct due to three factors: publications in journals and conferences in various areas, peer-review papers, and familiarity and facility with using the platforms. According to Stravinskienė and Serafinas [15], Scopus, WoS, and Elsevier are important multidisciplinary bibliographic databases with multiple sources.

TABLE III
SEARCH CHARACTERISTICS

Element	Details
Source	Scopus (https://www.scopus.com) Web of Science/Clarivate (https://www.webofscience.com) Science Direct/Elsevier (https://www.sciencedirect.com)
Search characteristics	Articles in academic journals or conference materials with a date range limit

The search strings (or search queries SQ) for this SLR were constructed based on our knowledge of the topic, a preliminary ad-hoc search about the area, and an analysis of the research questions (RQ) defined in Table II. We can define some terms and synonyms, abbreviations, and alternative spellings for each RQ. After some iterations, we concluded that it was important to consider the search strings (Table IV) in the title, keywords, abstract, and the entire publication text. We have designed the SQ with combinations of key terms and the boolean operators AND or OR. SMEs were included in some SQs, as explained in the next table.

We have performed the SLR with the support of Parsif.al. In this web platform, we found key aspects that were important to document our search for each phase defined in Kitchenham and Charters [71]. This web tool helps the researchers perform SLR and document the results.

TABLE IV
SEARCH STRINGS FOR PERFORMING SLR

SQ	Principal definition
SQ1	("Business Process Automation" OR “BPA”) AND ("adoption" OR "implementation") AND (“limitations” OR “challenges” OR “issues” OR “obstacle” OR “disadvantages”) AND "benefits"
SQ2	("Business Process Automation" OR “BPA”)
SQ3	("Business Process Automation" OR “BPA”) AND “automation technologies” AND ("adoption" or "implementation") AND (“guidelines” OR “framework” OR “roadmap”) AND “SME”
SQ4	("Business Process Management" OR “BPM”) AND ("Business Process Automation" or” BPA”)
SQ5	("Business Process Automation" OR “BPA” OR “automation technologies”) AND ("success criteria" OR “KPI”) AND “SME”
SQ6	("Business Process Automation" OR “BPA”) AND “automation technologies” AND ("adoption" or "implementation") AND (“guidelines” OR “framework” OR “roadmap”)
SQ7	(“Business Process Management” OR “Business Process Automation”) AND (“guidelines” OR “framework” OR “roadmap” or “model”)

C. INCLUSION, EXCLUSION, AND QUALITY CRITERIA

To accomplish the goals of this SLR, the focus of this search protocol was studies published between the years 2016 and 2023. The research team, paying attention to the volatility of the technology area, defined this period. For this literature review, we have the selection criteria listed in Table V.

TABLE V
EXCLUSION AND INCLUSION CRITERIA

Exclusion Criteria	Inclusion Criteria
E1 - Other languages than English	I1 - Peer-reviewed
E2 - Duplicated reference	I2 - Related to the topic and problem
E3 - Grey literature	I3 - Published in journals or conference proceedings
E4 - Publication year before 2016	I4 – Title/abstract related to BPM/BPA
E5 - Paper main focus out-of-scope	I5 - Title/abstract related to technologies that support BPM/BPA
E6 - Paper cannot be accessed	
E7 - Papers incomplete	
E8 - Position papers without conclusions	

After retrieving all the articles from SQ in data sources, and according to Kitchenham and Charters [71], the review protocol includes the classification of each one, following a checklist of parameters that guarantee a minimum quality of the paper. For this SLR, for the quality assessment of one paper, we have formulated a set of six questions (Table VI) with three possible answers, each with a weight (Yes – score 1; Partially – score 0.5; No – score 0).

The final quality assessment score is calculated by summing all scores of the six answers. The study has a score with a maximum value of 6 and a cut-off of less than 2.5, which means that the paper with a total score of 2.5 is included in this study. In this way, we can ensure that this study has a broad scope.

TABLE VI
QUESTIONS AND POSSIBLE ANSWERS WITH WEIGHT

Questions Quality Assessment (QA)		Answers (Weight)	
QA1: Is the scope/purpose of the study related to the question for this research?	Yes (1)	Partially (0.5)	No (0)
QA2: Is the methodology explained?	Yes (1)	Partially (0.5)	No (0)
QA3: Conclusions proved with experience?	Yes (1)	Partially (0.5)	No (0)
QA4: Has the study already been cited?	Yes (1)	Partially (0.5)	No (0)
QA5: Are there some considerations for answers in RQ?	Yes (1)	Partially (0.5)	No (0)
QA6: Can the study be considered as a primary one?	Yes (1)	Partially (0.5)	No (0)

D. RELEVANT PUBLICATION SELECTION PROCEDURE

Conducting is the second phase of the SLR methodology by Kitchenham and Charters [71]. The relevant publication is selected according to inclusion and exclusion criteria in this phase. Kitchenham and Charters [71] mentioned that a systematic review aims to find as many primary studies relating to the research question as possible. After selecting papers, we will proceed with data extraction, monitoring, and synthesis of findings.

The search queries were performed in Scopus, Web of Science (WoS), and Science Direct (SD) in the first quarter of 2023, taking into consideration filters that were representative of the E1 (*language* different from *English*), E3(I1/I3) (*peer-reviewed* and published in *conferences proceedings* or *journals*), and E4 (time frame between *2016 and 2023*). The final number of studies selected for this study was reduced by a process with successive exclusion filters (see Table VII).

We started with 25616 papers in the BibTeX file due to the search in all data sources. Then, it was imported into Parfif.al, and cleaned the duplicated references (*E2 - duplicated reference*). We found 7597 papers distributed in Table VII, a reduction of 70%.

After this process, we applied another exclusion criteria paper, the focus being *out-of-scope* (E5). This analysis revealed areas like *Chemical Sciences & Engineering, Environment Sciences & Engineering*, *Agricultural Engineering*, and *Biotechnology & Biology* where the BPA acronym appears. In some papers, BPA meant *bisphenol*; in others, it meant *basic probability assignment* or *backpropagation algorithm*, among other meanings, instead of *business process automation*. At the end of this step, we remained with 1381 papers (a reduction of 82%) related to business process management/business process automation, distributed as Table VII.

TABLE VII
PROCESS OF REDUCING THE NUMBER OF RELEVANT STUDIES PER DATA SOURCE AND CRITERIA

	Papers after criteria				
Criteria	Scopus	WoS	SD	Total	Reduction
E1/E3(I1/I3)/E4	12060	11349	2207	25616	
E2	6574	9987	1036	7597	70.34%
E5	938	318	125	1381	81.82%
E6/E7/E8	246	73	24	343	75.16%

The next step was reading the titles and at least the abstracts of these 1381 papers. In some cases, we had read the introduction as well. In this stage, we found some publications that *could not be accessed* (E6), others *incomplete* (E7), and others were *position papers without conclusions* (E8), reducing about 75% of the publications (Table VII). Ultimately, we passed to the next task, checking the quality assessment, 343 publications (about 25% of 1381 from the last step).

Kitchenham and Charters [71] considered is "*critical to assessing the quality*." The final quality assessment score is in Table VIII. When analyzing all the publications, we have found that some are in the scope of Business Process Management (BPM) but do not focus on the automation of the process, how to automate a business process, how to improve a business process, or technologies to do that. Others considered its goals in data mining techniques (DM) or how to achieve more successful results in DM. The articles referencing automating processes in the industry using sensors, IoT, and other technologies were discarded. Also, papers that have descriptions of commercial tools for automation were discarded. The quality assessment was performed by answering the QA1 to QA6 presented in Table VI. This study references 99 articles with a cuff-off of less than 2.5, with the average value being 3.7.

TABLE VIII
QUALITY ASSESSMENT SCORE

Score	Number of articles	Observations
0 – 2,49 (low quality)	244	Not selected for SLR
2,50 – 4,49 (medium quality)	78	Selected for SLR
4,50 – 6 (high quality	21	Selected for SLR
Total	343	

E. DATA EXTRACTION METHOD

For each of the 99 relevant publications, a method to extract data is performed to answer the SRL five questions defined in the first stage of Kitchenham and Charters [71]. Using Parsif.al tool, the authors have extracted the described data from Table IX:

TABLE IX
LIST OF THE EXTRACTED DATA FOR THIS SLR

Data	Description	RQ or Study Data
Title	Title of the study	Study data
Authors	Authors of the study	Study data
Year	Year of the study	Study data
DOI	Digital Object Identifier	Study data
Journal	Where was the publication of the study	Study data
Publisher	Publisher of the study	Study data
Research question	Which RQ the study answers	RQ
Process attributes to be eligible for automation	How to typify the processes for automation?	RQ1
Technologies/tools used for automation or improvements	What are the main technologies/tools to support business process automation?	RQ2
Technologies characteristics that define them	What are the main technologies/tools to support business process automation?	RQ2
Negative influencing factors for BPA in SMEs	What are the main barriers, risks, and limitations in business process automation in SMEs?	RQ3
Positive influencing factors for BPA in SMEs	What are the main barriers, risks, and limitations in business process automation in SMEs?	RQ3

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Data	Description	RQ or Study Data
Evaluation of the success of automation in SMEs	How to evaluate the success of business process automation in SMEs?	RQ4
Frameworks for BPA in SMEs	What frameworks exist to support process automation for SMEs?	RQ5
Guidelines for BPA in SMEs	What frameworks exist to support process automation for SMEs?	RQ5
Roadmap for BPA in SMEs	What frameworks exist to support process automation for SMEs?	RQ5
Methodology for BPA in SMEs	What frameworks exist to support process automation for SMEs?	RQ5

F. STUDIES OVERVIEW

The explained selection process (section Relevant Publication Selection Procedure) resulted in a reduction of papers from the initial 25616 studies from the searches (Table VII) to 99 references that were identified as a publication of interest for this SLR (Table X). In Table X, there is information about the title, the authors, the year of publication, and the quality assessment score.

Most of them have a final score of the quality assessment between 2.5 and 4.49 (Table VIII) and a mean score of 3.7 (Figure 2).

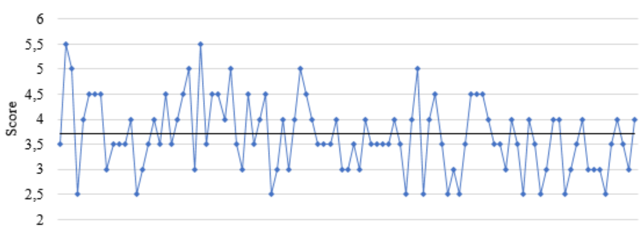

FIGURE 2. **Score of the 99 papers selected from the quality assessment (mean at value 3.7).**

1) PAPER YEAR OF PUBLICATION ANALYSIS

From Table X, we can analyze the year of publication. Table XI presents the papers selected and distributed per year. Since we performed the search in the first quarter of 2023, it is normal that we do not have many references to that year.

TABLE XI
PAPERS BY PUBLICATION YEAR

Year	Number of papers
2016	6
2017	5
2018	11
2019	18
2020	22
2021	13
2022	22
2023	2
Total	99

As demonstrated in Figure 3, four years contributed to more than 75% of the articles for this SLR. From 2019 to 2022, we have selected 75 papers (about 75,76%). The remaining papers were distributed as 6 in 2016, 5 in 2017, 11 in 2018, and 2 in 2023.

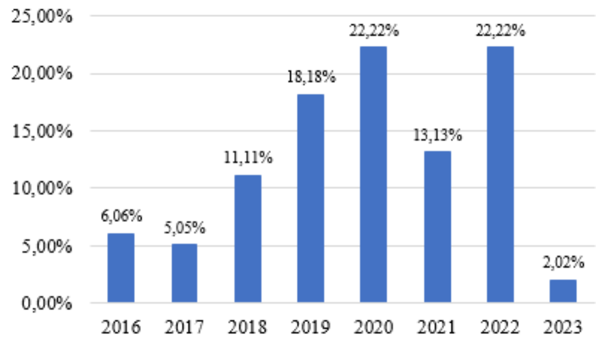

FIGURE 3. **% of papers per year of publication.**

2) ORIGIN OF PAPER ANALYSIS

Regarding the Journal or Conference where the paper was published, we found that there were a large of different ones. This fact supports the idea that business process automation, besides international (Figure 4), is a hot topic in diverse areas like services, commerce, accounting, and insurance, among others. This information was retrieved in July 2023 from the website *scimagojr.com*. Two notes from the nationalities of the publications: the United States is the leader, and there are 9 journals or conferences where nationality was not found in *scimagojr.com*.

To better understand the origin of publications, the authors have divided them into *Journals* and *Book Series* (Table XII) and another group of conferences and *proceedings* (Table XIII). Figure 5 notes that 67% of the papers used in this SLR are from *Journals* and *Book Series*.

Visiting Figure 4 and performing the division of the papers by the two groups formed, the authors show in Figure 6 the number of papers by nationality and group of publication.

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TABLE X
RESUME OF THE PAPERS IN THIS SLR

Ref	Authors	YEAR	Title	Score
[47]	Herm et al.	2022	A framework for implementing robotic process automation projects	5,5
[72]	Dumčius and Skersys	2019	Improvement and digitalization of business processes in small-medium enterprises	5,5
[46]	Eulerich et al.	2022	A Framework for Using Robotic Process Automation for Audit Tasks	5,0
[39]	Noppen et al.	2020	How to Keep RPA Maintainable?	5,0
[53]	Liutkevičienė et al.	2022	Leveraging capabilities for digitally supported process improvement: a framework for combining Lean and ERP	5,0
[73]	Sobczak	2021	Robotic Process Automation implementation, deployment approaches, and success factors – an empirical study	5,0
[74]	Siderska	2021	The Adoption of Robotic Process Automation Technology to Ensure Business Processes during the COVID-19 Pandemic	5,0
[56]	Neto et al.	2022	AKIP Process Automation Platform: A Framework for the Development of Process-Aware Web Applications	4,5
[11]	Seymour et al.	2019	Applied business process management: An information systems approach to improve service delivery in public hospitals of low- and middle-income countries	4,5
[44]	Huang and Vasarhelyi	2019	Applying robotic process automation (RPA) in auditing: A framework	4,5
[40]	Ludacka et al.	2019	Global accounting with the TIM BPM Suite at Deutsche Bahn Group	4,5
[8]	Denner et al.	2018	How to Exploit the Digitalization Potential of Business Processes	4,5
[16]	Alves et al.	2018	Integrating requirements and business process models in BPM projects	4,5
[38]	König et al.	2020	Integrating Robotic Process Automation into Business Process Management	4,5
[13]	Szelągowski	2021	Practical assessment of the nature of business processes	4,5
[51]	Smirnov et al.	2022	Product Configuration Automation: Digital Transformation Platform and Case Study	4,5
[75]	Flechsig et al.	2021	Robotic Process Automation in purchasing and supply management: A multiple case study on potentials, barriers, and implementation	4,5
[76]	Patri	2020	Robotic Process Automation: Challenges and Solutions for the Banking Sector	4,5
[77]	Leite et al.	2020	Process Automation and Blockchain in Intelligence and Investigation Units: An Approach	4,5
[78]	Zhang et al.	2021	Robotic Process Automation (RPA) Implementation Case Studies in Accounting: A Beginning to End Perspective	4,5
[79]	Sobczak	2022	Robotic Process Automation as a Digital Transformation Tool for Increasing Organizational Resilience in Polish Enterprises	4,5
[45]	Feio and dos Santos	2022	A Strategic Model and Framework for Intelligent Process Automation	4,0
[27]	Nurhayati and Fitrisari	2016	Business Process Automation of Document Filing Based Alfresco for Referral BPJS Patient (Case Study: BPJS Center of Dharmais Cancer Hospital)	4,0
[54]	Noureddine and Oualid	2018	Extraction of ERP Selection Criteria using Critical Decisions Analysis	4,0
[19]	Mendling et al.	2018	How do Machine Learning, Robotic Process Automation, and Blockchains Affect the Human Factor in Business Process Management?	4,0
[37]	Torkhani et al.	2018	Intelligent Framework For Business Process Automation And Re-engineering	4,0
[6]	Athanasopoulos et al.	2017	Process Management in the Greek Public Sector: A case study in the Municipality of Kalamaria	4,0
[20]	Ahmad and Van Looy	2019	Reviewing the historical link between Business Process Management and IT: making the case towards digital innovation	4,0
[31]	Wewerka and Reichert	2021	Robotic process automation - a systematic mapping study and classification framework	4,0
[61]	Costa et al.	2022	Robotic Process Automation (RPA) adoption: a systematic literature review	4,0
[35]	Hartley and Sawaya	2019	Tortoise, not the hare: Digital transformation of supply chain business processes	4,0
[64]	Sobczak	2019	Developing a Robotic Process Automation Management Model	4,0
[80]	Asquith and Horsman	2019	Let the robots do it! – Taking a look at Robotic Process Automation and its potential application in digital forensics	4,0
[81]	Rizk et al.	2020	A Conversational Digital Assistant for Intelligent Process Automation	4,0
[82]	Choi et al.	2021	Candidate Digital Tasks Selection Methodology for Automation with Robotic Process Automation	4,0
[83]	Mishra et al.	2019	People & Process Dimensions of Automation in the Business Process Management Industry	4,0
[84]	Wellmann et al.	2020	A Framework to Evaluate the Viability of Robotic Process Automation for Business Process Activities	4,0
[85]	Aguirre and Rodriguez	2017	Automation of a Business Process Using Robotic Process Automation (RPA): A Case Study	4,0
[86]	Reitsma and Hilletofth	2018	Critical success factors for ERP system implementation: a user perspective	4,0
[66]	Kouriati et al.	2020	Critical Success Factors on the Implementation of ERP Systems: Building a Theoretical Framework	4,0
[87]	Rocha et al.	2017	In the process babel: Definitions, concepts, and tools in a disordered field	4,0
[68]	Plattfaut et al.	2022	The Critical Success Factors for Robotic Process Automation	4,0
[69]	Santos et al.	2020	Towards Robotic Process Automation implementation: An end-to-end perspective	4,0
[59]	Vanwersch et al.	2016	A critical evaluation and framework of business process improvement methods	3,5
[88]	Leon	2020	Framework for the Automation of Business Processes	3,5
[36]	Di Ciccio et al.	2019	Blockchain Support for Collaborative Business Processes	3,5
[23]	Lazareva et al.	2022	Business Process Automation in Retail	3,5
[41]	Martinek-Jaguszewska and Rogowski	2022	Development and Validation of the Business Process Automation Maturity Model: Results of the Delphi Study	3,5
[89]	Volik	2019	Features of the analysis of business processes of the company (on the example of customer service in a travel agency)	3,5
[12]	Caforio et al.	2020	HINT project: a BPM teleconsultation and telemonitoring platform	3,5
[90]	Wiradinata	2018	Information systems adoption among SMEs in developing country: The case of Gerbang Kertasusila	3,5

Ref	Authors	YEAR	Title	Score
[52]	E.R. et al.	2019	Model for BPM implementation assessment: evidence from companies in Indonesia	3,5
[15]	Stravinskienė and Serafinas	2021	Process Management and Robotic Process Automation: The Insights from Systematic Literature Review	3,5
[25]	Susilo et al.	2021	The Implementation of Robotic Process Automation for Banking Sector Case Study of A Private Bank in Indonesia	3,5
[14]	Šperka and Halaška	2022	The performance assessment framework (PPAFR) for RPA implementation in a loan application process using process mining	3,5
[91]	Sobczak and Ziora	2021	The Use of Robotic Process Automation (RPA) as an Element of Smart City Implementation: A Case Study of Electricity Billing Document Management at Bydgoszcz City Hall	3,5
[62]	Syed et al.	2020	Robotic Process Automation: Contemporary themes and challenges	3,5
[65]	Hindel et al.	2020	Robotic Process Automation: Hype or Hope?	3,5
[92]	Madakam et al.	2019	The future digital work force: Robotic Process Automation (RPA)	3,5
[93]	Romão et al.	2019	Robotic Process Automation: A case study in the Banking Industry	3,5
[94]	Siderska	2020	Robotic Process Automation - a driver of digital transformation?	3,5
[95]	Reddy et al.	2019	A Study of Robotic Process Automation Among Artificial Intelligence	3,5
[96]	Vitharanage et al.	2020	An Empirically Supported Conceptualization of Robotic Process Automation (RPA) Benefits	3,5
[97]	Sethi et al.	2020	Embedding Robotic Process Automation into Process Management: Case Study of using task	3,5
[48]	Enríquez et al.	2020	Robotic Process Automation: A Scientific and Industrial Systematic Mapping Study	3,5
[58]	Moreira et al.	2023	Business Process Automation in SMEs	3,5
[98]	Reungyu and Waiyanet	2022	An Exploratory Study on the Impact of RPA (Robotic Process Automation) Implementation on Behavioral Attitudes and Intentions within Organizations	3,5
[99]	Katsonis and Sfakianakis	2016	Business process automation in Greek telecommunications companies and marketing strategies	3,5
[67]	Zayas-Cabán et al.	2021	Identifying Opportunities for Workflow Automation in Health Care: Lessons Learned from Other Industries	3,5
[100]	Aboumasoudi et al.	2021	Studying the critical success factors of ERP in the banking sector: a DEMATEL approach	3,5
[101]	Wewerka and Reichert	2020	Towards Quantifying the Effects of Robotic Process Automation	3,5
[43]	Singh, V.	2022	Automation of Business Process Using RPA (Robotic Process Automation)	3,0
[55]	Waszkowski et al.	2017	Customer Service Processes Automation in Administrative Office with RFID Tagged Documents	3,0
[102]	Tarutė et al.	2018	Identifying factors affecting the digital transformation of SMEs	3,0
[21]	Gabryelczyk et al.	2022	Motivations to Adopt BPM in View of Digital Transformation	3,0
[50]	Hull and Nezhad	2016	Rethinking BPM in a Cognitive World: Transforming How We Learn and Perform Business Processes	3,0
[49]	Hofmann et al.	2020	Robotic process automation	3,0
[10]	van der Aalst et al.	2018	Views on the Past, Present, and Future of Business and Information Systems Engineering	3,0
[26]	Chakraborti et al.	2020	From Robotic Process Automation to Intelligent Process Automation – Emerging Trends	3,0
[63]	Ng et al.	2021	A systematic literature review on intelligent automation: Aligning concepts from theory, practice, and future perspectives.	3,0
[103]	Puica	2022	How Is it a Benefit to Use Robotic Process Automation in Supply Chain Management?	3,0
[104]	Phaphoom et al.	2018	A Combined Method for Analysing Critical Success Factors on ERP Implementation	3,0
[105]	Chatzoglou et al.	2016	Critical success factors for ERP implementation in SMEs Prodromos	3,0
[106]	Van Looy	2022	Employees' attitudes towards intelligent robots: a dilemma analysis	3,0
[107]	van der Aalst	2020	On the Pareto principle in process mining, task mining, and robotic process automation	3,0
[57]	Flechsig et al.	2022	Towards an Integrated Platform for Business Process Management Systems and Robotic Process Automation	3,0
[108]	Wanner et al.	2019	Process selection in RPA projects - Towards a quantifiable method of decision-making	3,0
[109]	Seymour and Koopman	2022	Analysing factors impacting BPMS performance: a case of a challenged technology adoption	3,0
[60]	Moreira et al.	2023	Process automation using RPA – a literature review	2,5
[42]	Engel et al.	2022	Cognitive automation.	2,5
[7]	Laar and Seymour	2017	Redesigning business processes for small and medium enterprises in developing countries	2,5
[110]	Ansari et al.	2019	A Review on Robotic Process Automation - The future of Business Organizations	2,5
[111]	Yatskiv et al.	2020	Method of Robotic Process Automation in Software Testing Using Artificial Intelligence	2,5
[112]	Dechamma and Shobha	2020	A Review on Robotic Process Automation	2,5
[113]	Dey and Das	2019	Robotic process automation: assessment of the technology for transformation of business processes	2,5
[114]	Sharma et al.	2022	Applications, Tools, and Technologies of Robotic Process Automation in Various Industries	2,5
[115]	van der Aalst et al.	2016	Business Process Management: Don't Forget to Improve the Process!	2,5
[116]	Kukharenko and Mankov	2022	Digitalization of the Company's Business Processes Based on Technological Innovations	2,5
[117]	van der Aalst et al.	2018	Robotic Process Automation	2,5

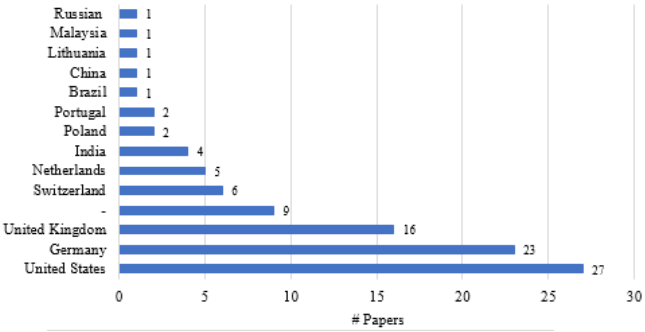

FIGURE 4. **Number of papers by publications nationalities.**

In Table XII, column *Q* represents the *quartile* of the publication, and # represents the *number of papers* selected from the Journal or Book Series. This table shows the list of journals and book series arranged by type (journal first, book series are ticked with (*BS*)), then by the score of quartiles, and finally alphabetically by the name of the publication. Besides cited information, Table XII included two more columns: *H-Index*[1] and *SJR*[2]2022.

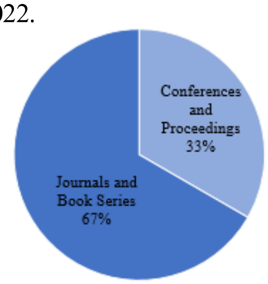

FIGURE 5. **% of papers divided into two groups.**

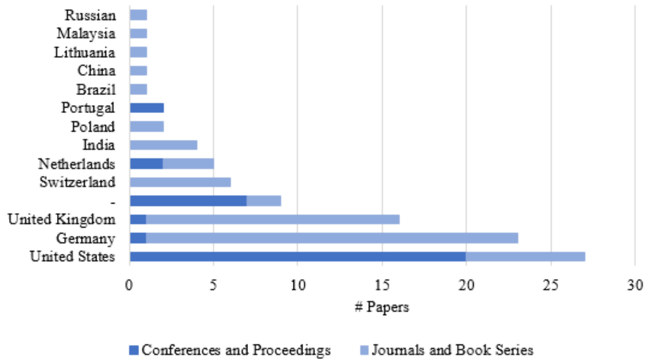

FIGURE 6. **Number of papers by publications, nationalities, and groups.**

In the Conferences and Proceedings group, the authors did not see the relevance of mentioning column Q because all values were zero. Table XIII shows the list of conferences and proceedings alphabetically by the name of the publication.

Table XIII also included two more columns: *H-Index* and *SJR2022*.

Making a detailed overview of the origin of publications, regarding the *quartile* of each one, the authors inform that 28 papers were published in Q1 Journals/Book Series, 17 in Q2 Journals/Book Series, 7 in Q2 Journals/Book Series, 2 in Q4 Journals/Book Series, 12 not classified with quartile despite belonging to Journals/Book Series, and 33 papers from Conferences and Proceedings (Figure 7). This study has 54 (about 54.55%) papers classified in the Journals/Book Series with quartile (Figure 8).

TABLE XII
SELECTED PAPERS, JOURNALS, AND BOOK SERIES

Journal or Book Series	Country	H-Index	SJR2022	Q	#
Accounting Horizons	United States	87	1.00	Q1	1
Advanced Engineering Informatics	United Kingdom	99	1.71	Q1	1
Applied Clinical Informatics	Germany	37	0.97	Q1	1
Business and Information Systems Engineering	Germany	57	1.54	Q1	5
Business Horizons	United Kingdom	106	2.48	Q1	1
Business Process Management Journal	United Kingdom	90	0.92	Q1	3
Computers in Industry	Netherlands	117	2.65	Q1	2
Contemporary Accounting Research	United States	115	3.01	Q1	1
Electronic Markets	Germany	49	1.55	Q1	2
Energies	Switzerland	132	0.63	Q1	1
Enterprise Information Systems	United Kingdom	54	1.04	Q1	1
European Business Review	United Kingdom	51	3.67	Q1	1
IEEE Access	United States	204	0.93	Q1	1
Information Systems Management	United Kingdom	62	1.16	Q1	2
International Journal of Accounting Information Systems	United States	60	1.16	Q1	1
Journal of Purchasing and Supply Management	United Kingdom	95	2.09	Q1	1
Sustainability	Switzerland	136	0.66	Q1	3
Applied Sciences	Switzerland	101	0.49	Q2	1
Communications of the Association for Information Systems	United States	56	0.64	Q2	1

[1] H-Index is related to the number of articles (h) that have received also at least h citations. Quantifies journal scientific productivity and scientific impact and applies to scientists, and countries, among other items (https://www.scimagojr.com/help.php)

[2] SJR (SCImago Journal Rank) Indicator informs the average number of weighted citations received in the selected year relative to the number of publications of the journal in the three previous years (https://www.scimagojr.com/help.php)

Journal or Book Series	Country	H-Index	SJR2022	Q	#
Engineering Management in Production and Services	Poland	17	0.37	Q2	2
Information Systems and e-Business Management	Germany	43	0.71	Q2	4
International Journal of Procurement Management	Switzerland	24	0.39	Q2	1
Knowledge and Process Management	United Kingdom	49	0.57	Q2	1
Lecture Notes in Business Information Processing (BS)	Germany	56	0.35	Q2	6
Software and Systems Modeling	Germany	55	0.7	Q2	1
Business Informatics	Russian Federation	10	0.23	Q3	1
Electronic Journal of Information Systems in Developing Countries	China	25	0.46	Q3	1
Forensic Science International: Reports	Netherlands	5	0.33	Q3	1
Informatik-Spektrum	Germany	21	0.26	Q3	1
International Journal of Advanced Computer Science and Applications	United Kingdom	35	0.26	Q3	2
International Journal of Electronic Customer Relationship Management	United Kingdom	16	0.35	Q3	1
International Journal of Business Process Integration and Management	United Kingdom	19	0.16	Q4	1
Communications in Computer and Information Science (BS)	Germany	62	0.19	Q4	1
AIS Transactions on Enterprise Systems	Germany	-	-	-	1
Entrepreneurship and Sustainability Issues	Lithuania	36	-	-	1
International Journal for Research in Applied Science & Engineering Technology	-	-	-	-	1
International Journal of Engineering and Advanced Technology	India	25	-	-	1
International Journal of Management	India	-	-	-	1
International Journal of Research in Engineering, Science and Management	India	-	-	-	1
International Journal of Scientific and Research Publications	India	-	-	-	1
International Journal of Systems Engineering	United States	-	-	-	1
Journal of Information Systems and Technology Management	Brazil	-	-	-	1
Journal of Supply Chain and Customer Relationship Management	United States	-	-	-	1
Journal of Telecommunication, Electronic and Computer Engineering	Malaysia	23	-	-	1
Management of Organizations: Systematic Research	-	-	-	-	1

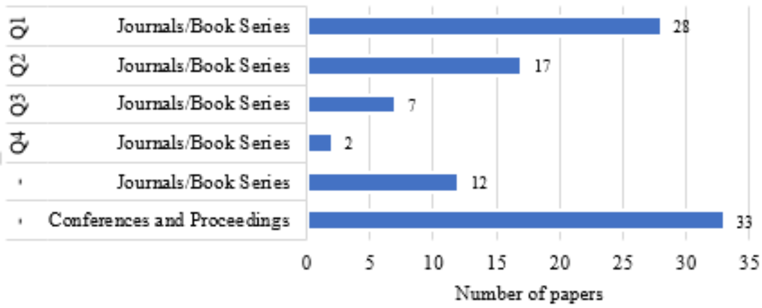

FIGURE 7. **Count of papers per quartile.**

TABLE XIII
SELECTED CONFERENCES AND PROCEEDINGS

Conference and Proceedings	Country	H-Index	SJR2022	#
ACM International Conference Proceeding Series	United States	137	0.21	4
Business Process Management. Lecture Notes in Computer Science	--	-	-	2
CEUR Workshop Proceedings	United States	62	0.2	2
Conference on Software Engineering and Advanced Applications	United States	13	-	1
European Conference on Information Systems	-	-	-	1
Federated Conference on Computer Science and Information Systems	United States	16	-	1
Iberian Conference on Information Systems and Technologies	United States	20	-	2
IEEE International Enterprise Distributed Object Computing Workshop	United States	36	-	1
International Conference on Advanced Computer Information Technologies	United States	8	-	1
International Conference on Advances in Science & Technology	-	-	-	1
International Conference on Business and Industrial Research	United States	8	-	1
International Conference on Data Science, Technology and Applications	Portugal	5	-	1
International Conference on Decision Aid Sciences and Applications	-	-	-	1
International Conference on Informatics and Computing	United States	10	-	1
International Conference on Information Systems	United States	23	-	1
International Conference on Innovative Intelligent Industrial Production and Logistics	-	-	-	1
International Conference on Intelligent Systems	-	-	-	1
International Conference on Management Leadership and Governance	United Kingdom	3	-	1
International Conference on Mathematics and Computers in Sciences and Industry	United States	4	-	1
International Conference on Quality Management, Transport and Inf. Security, Inf. Technologies	United States	11	-	1

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Conference and Proceedings	Country	H-Index	SJR2022	#
International Conference on Research Challenges in Information Sciences	United States	9	-	1
International Conference on Web Information Systems and Technologies	Portugal	5	-	1
International Conference on Wirtschaftsinformatik	Germany	-	-	1
International Joint Conference on Computer Science and Software Engineering	United States	8	-	1
International Scientific Conf. on Inf. Tec. and Manag. Science of Riga Technical University	United States	3	-	1
Procedia Computer Science	Netherlands	109	0.51	1
Proceedings of the International Conference on Electronic Business	Netherlands	10	-	1

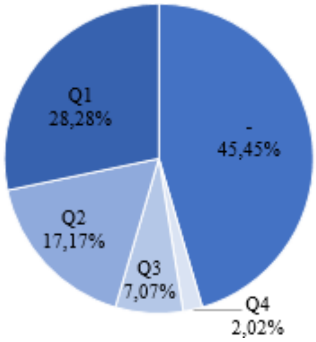

FIGURE 8. **% of papers per quartile.**

Making a comparison with the six (limit imposed by *scimagojr.com* tool) most cited Q1 Journals in this study - *Lecture Notes in Business Information Processing* with 6 papers, *Business and Information Systems Engineering* with 5 papers, *Information Systems and e-Business Management* with 4 papers, *Business Process Management Journal* with 3 papers, *Sustainability* with 3 papers and *Computers in Industry* with 2 papers, the Journal with more elevated H-Index value is *Sustainability* with 136, and the bottom place is for the *Information Systems and e-Business Management* with H-Index of 43. The compare journal tool of *scimagojr.com* makes a relation of the quartile of the citing and cited journals, which we can observe in Figure 9. This group of six Q1 Journals was cited by 42.96% of Q2 and cited in 66.32% of Q1 (Figure 9)

These six Journals have an interesting citation network (Figure 10). There is evidence of other Journals cited in this work.

However, regarding the origins of the publication of the set of 99 papers selected for this SLR, 17 papers belong to unlisted H-Index publications, and 82 have filiation in scored H-Index publications, which can be analyzed. There are 56 different publications origins with H-Index defined (Figure 11): 20 Conferences and Proceedings (about 36%) and 36 Journals/Book Series (about 64%).

The H-Index mean value is 48.73, and the H-Index weighted arithmetic mean is 55.99 (H-Index for all the origin of 82 papers with H-Index defined). The maximum value of the H-Index is 204, corresponding to Journal *IEEE Access* (Q1 – Computer Science). The minimum value of the H-Index is 3, belonging to two Conferences and Proceedings: The *International Conference on Management Leadership and Governance* and the *International Scientific Conference on Information Technology and Management Science of Riga Technical University*.

Regarding the SJR2022 index, the number of publications rated decreased concerning the 56 publications with a defined H-Index. 36 is the number of publications with an SJR 2022 defined (Figure 12). The Journals and Book Series group represents 91.67% (33 publications), and the remaining 3 publications are Conferences and Proceedings. These 36 different origins correspond to 61 papers used in this SLR.

FIGURE 9. **Quartile of the citing and cited journals (retrieved from scimagojr.com).**

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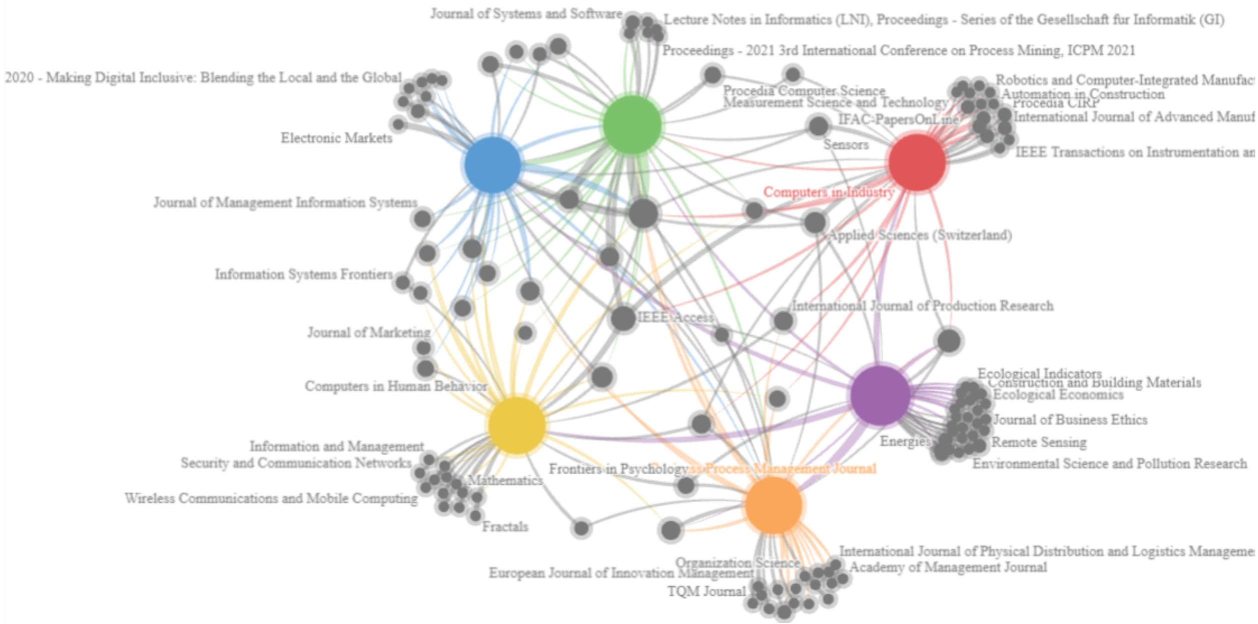

FIGURE 10. Citation network for the six more cited Q1 Journals in this SLR (retrieved from scimagojr.com).

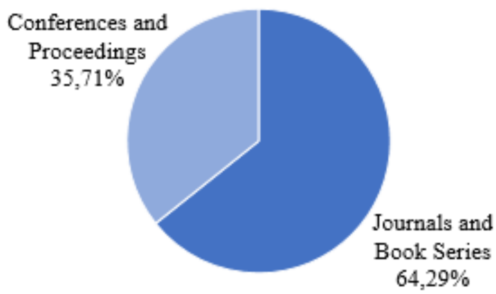

FIGURE 11. % of H-Index scored publications.

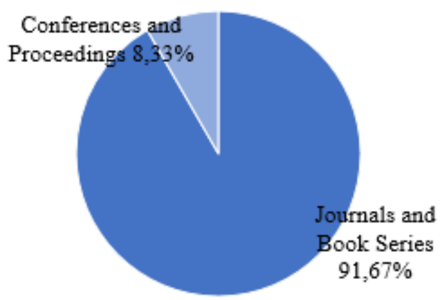

FIGURE 12. % of SJR 2022 scored publications.

The SJR 2022 mean value for this set of 36 publications is 0.96, and the SJR 2022 weighted arithmetic mean is 0.89 (SJR 2022 for all the origin of 61 papers with this value defined). The maximum value of the SJR 2022 is 3.67, corresponding to the Journal *European Business Review* (Q1 – Business and International Management). The minimum value of the SJR 2022 is 0.16, belonging to the Journal *International Journal of Business Process Integration and Management*.

Observing both H-Index and SJR 2022 values and considering publications that have defined both indicators (36 origins – 3 Conferences and Proceedings, and 33 Journals and Book Series), we can say that there exists much variation in this sample of publications (Figure 13).

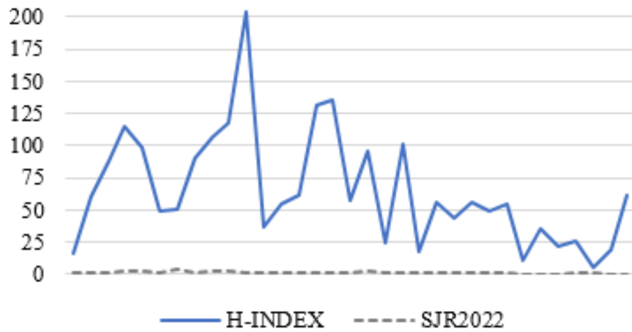

FIGURE 13. Variation of the values of H-Index and SJR2022 for origins that have booth indicators defined.

3) KEYWORDS CO-OCCURRENCE ANALYSIS

With the help of the VOSviewer tool, the authors performed an analysis of the keywords co-occurrence in the papers' texts. This analysis allowed us to verify the words that are related to each other and their respective link strength. For this review, the authors considered the title and abstract field, full counting, with the minimum number of occurrences of a term of 10. With these parameters, VOSviewer detected 1892 terms, and 48 meet the threshold. The relevance of terms was considered at 60%, leading to 29 terms being selected and grouped into 4 clusters (Figure 14). Cluster 1 has 10 items, Cluster 2 has 9, Cluster 3 has 8, and Cluster 4 has 2. The links between the items are 40, with a total link strength of 4406.

The top five keywords most cited were "*RPA,*" with a total link strength of 1180 and belonging to cluster 1; "*process automation,*" with a total link strength of 1144 and belonging to cluster 1; "*company*," with a total link strength of 435 and belongs to cluster 3, "*approach*" with a total link strength of 496 and belongs to cluster 2, and "*implementation*" with a total link strength of 517 and belongs to cluster 3 (see the top 10 in Table XIV). In the VOSviewer, the authors have selected "*process automation*" (Figure 15) and "*business process automation*" (Figure 16), as we can see in the next two network representations.

TABLE XIV
DESCRIPTION OF TOP 10 OCCURRENCE KEYWORDS

Keyword	Cluster	Occurrence	Link Strength
rpa	1	125	1180
process automation	1	119	1144
company	3	50	435
approach	2	46	496
implementation	3	44	517
task	1	43	365
framework	3	41	383
benefit	1	30	351
adoption	2	27	275
application	1	25	279

4) CO-AUTHORSHIP PRESENTATION

With the help of the VOSviewer tool, the authors performed the network related to the collaborations of the authors in the papers that were selected for this SLR. For this representation, the authors considered full counting, including all the papers (with many authors), with the minimum number of documents of an author of 1. With these parameters, VOSviewer detected 299 authors that meet the threshold. For each of the 200 authors, the total strength of co-authorship links with other authors was calculated. This resulted in 77 clusters, with the larger one having 22 authors (Figure 17). The links between the authors are 735, with a total link strength of 763. Through VOSviewer, the authors performed a drill down in detail in the large group of co-authorship (Figure 18). The group has the following clusters: 1 (red) with the author *van der Aalst, Will M. P.* (4 documents and total link strength of 24), 2 (green) with the author *Mendling, Jan* (4 documents and total link strength of 41); 3 (blue) with the author *Reijers, Hajo A.* (4 documents and total link strength of 25), and cluster 9 (purple) with no relevant author.

Table XV shows the top 10 authors, listed by the number of papers selected for this SLR, followed by the link strength.

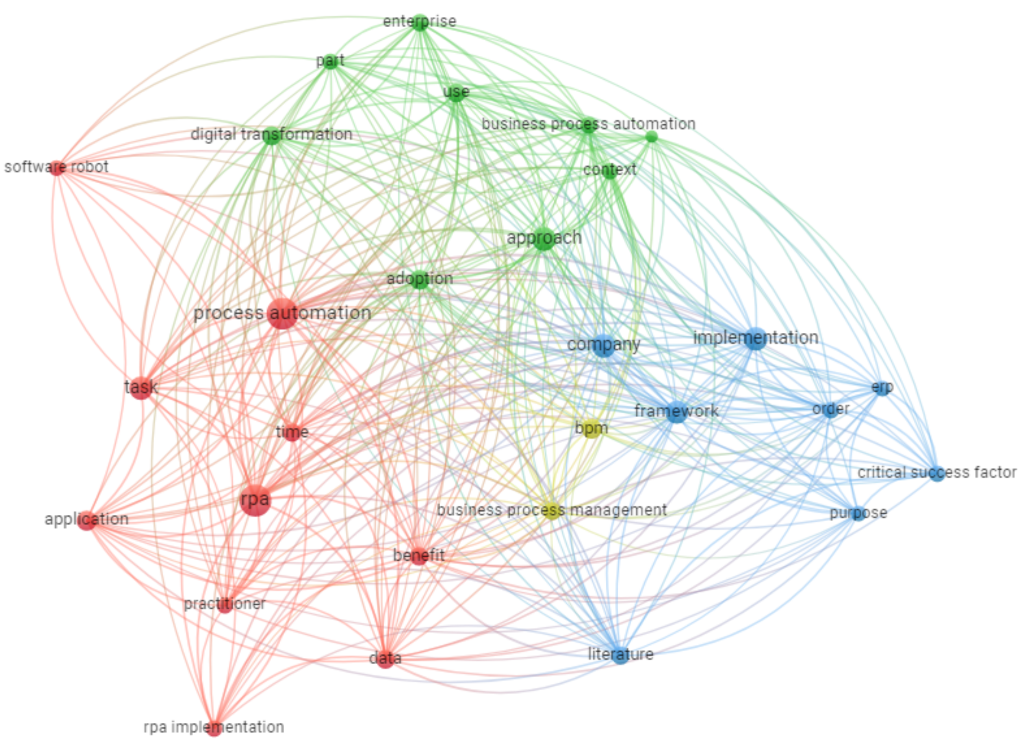

FIGURE 14. Keywords co-occurrence network visualization based on the number of occurrences.

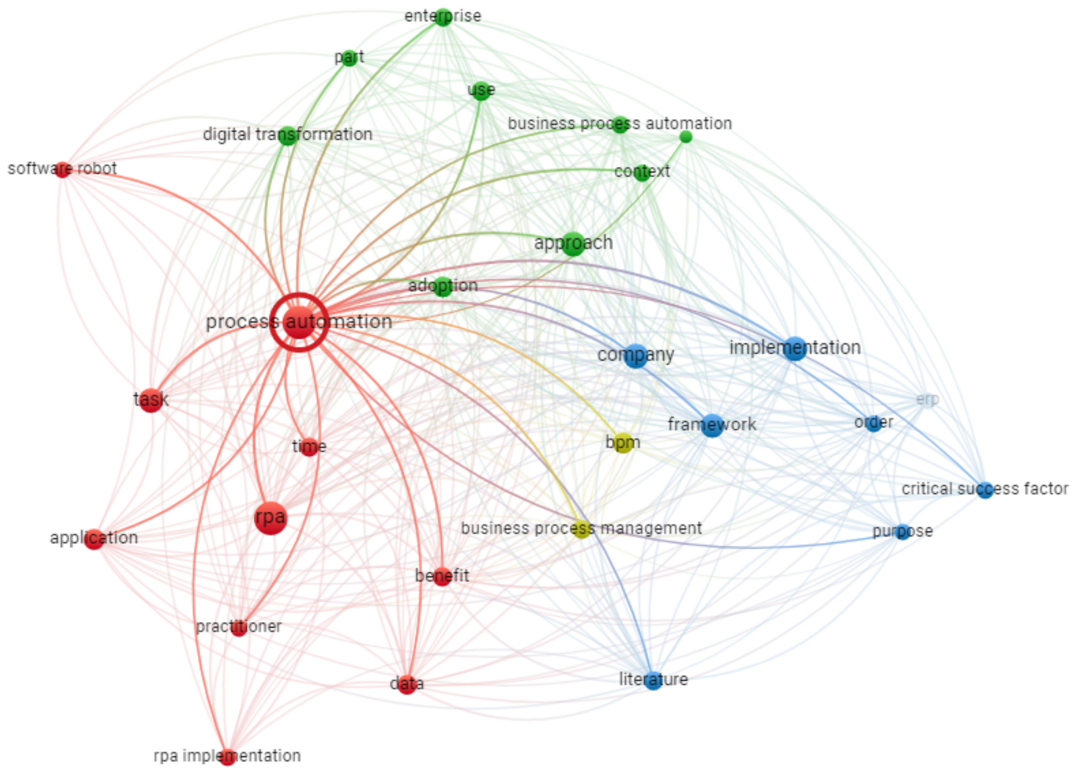

FIGURE 15. Network of "process automation".

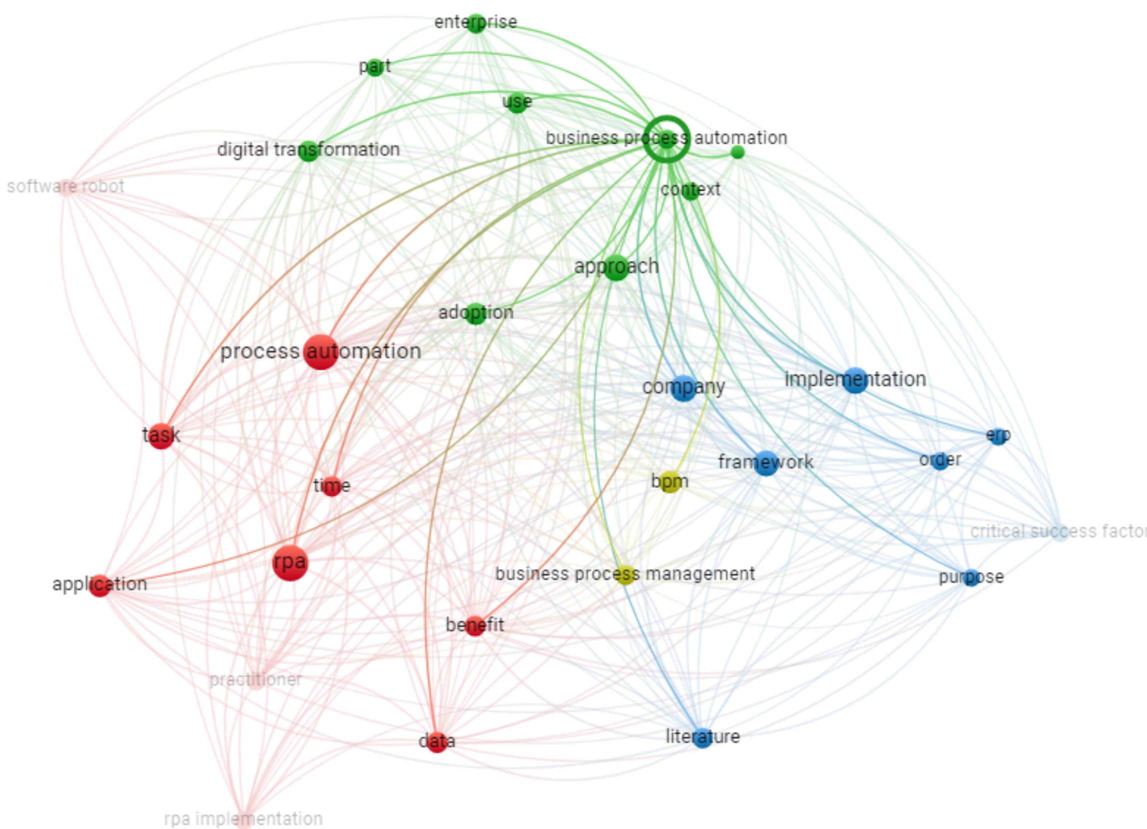

FIGURE 16. Network of "business process automation".

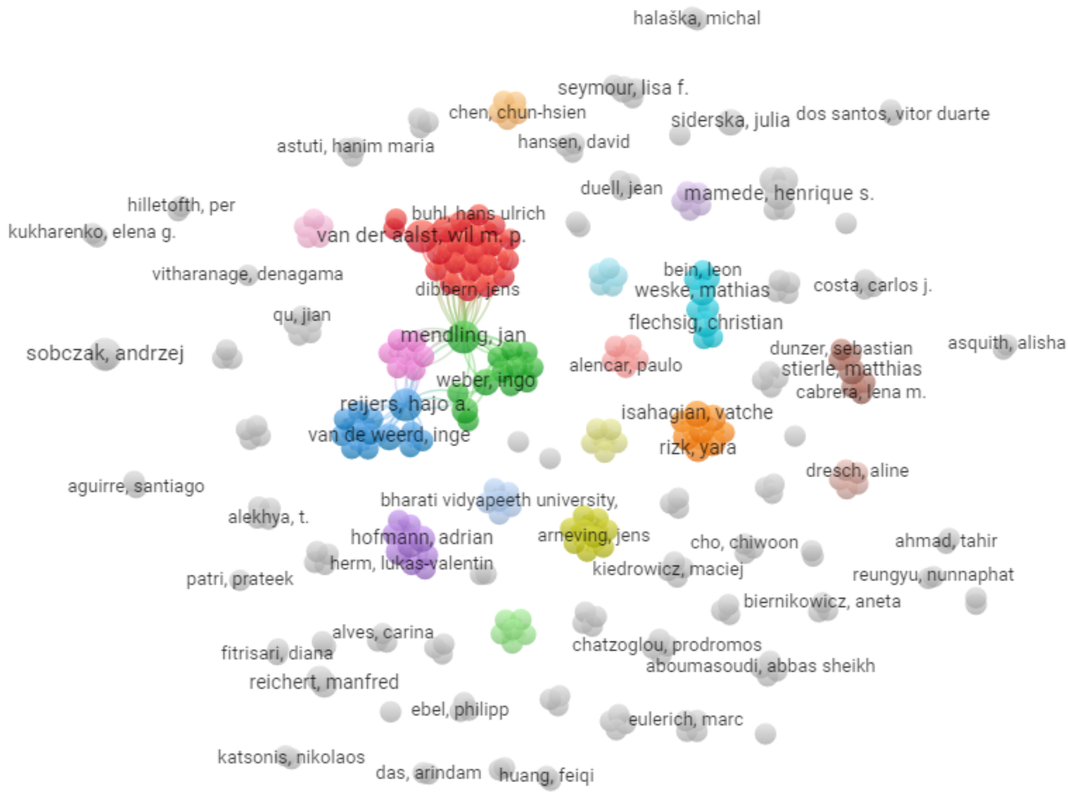

FIGURE 17. Co-authorship network visualization.

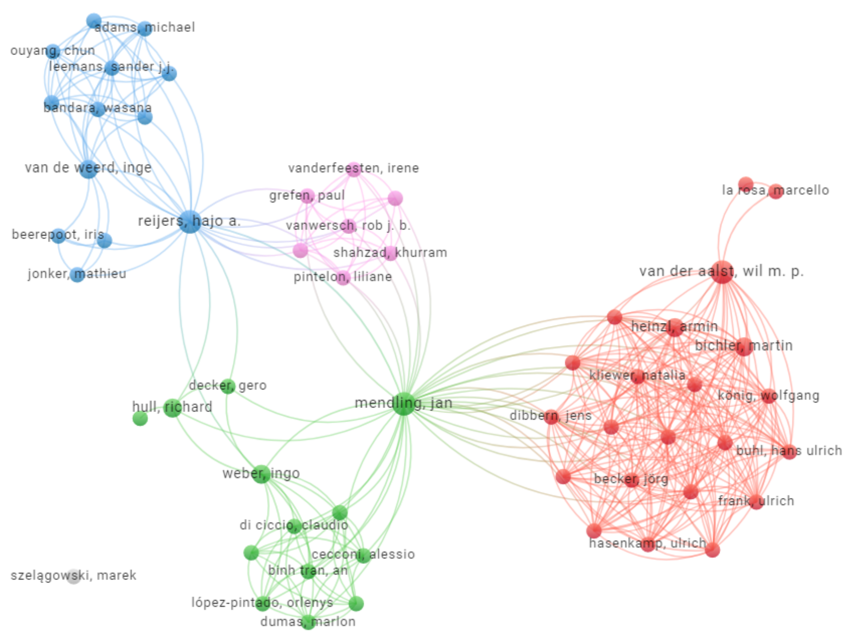

FIGURE 18. Detail of the large group of co-authorship.

TABLE XV
TOP 10 OF THE AUTHORS WITH THE NUMBER OF PAPERS AND LINK STRENGTH

Author	Papers	Link Strength
mendling, jan	4	41
reijers, hajo a.	4	25
van der aalst, wil m. p.	4	24
sobczak, andrzej	4	1
mamede, henrique s.	3	6
bichler, martin	2	22
heinzl, armin	2	22
van de weerd, inge	2	13
weber, ingo	2	13
isahagian, vatche	2	12

III. RESULTS AND DISCUSSION

In this chapter, the authors analyze the 99 papers selected in the quality assessment phase for this SLR to answer the research questions described in section II.A.

A. SELECTED PAPERS CONTRIBUTIONS

Besides the year, the title, the authors, the Journal/Conference of the publication, quality assessment, complete bibliographic reference for citation, and related to which RQ of this study was collected. Table XVI shows all the selected papers we used and their contribution to answering each of the five research questions that guided this SLR.

TABLE XVI
CONTRIBUTIONS OF EACH PAPER FOR THE ANSWERS TO THE FIVE RQS

REF.	Q1	Q2	Q3	Q4	Q5	REF.	Q1	Q2	Q3	Q4	Q5	REF.	Q1	Q2	Q3	Q4	Q5
[6]			X		X	[50]		X	X		X	[85]	X	X		X	
[7]			X			[51]		X			X	[86]		X	X	X	
[8]		X	X		X	[52]		X	X	X		[87]			X	X	
[10]	X	X		X		[53]		X		X	X	[88]			X		X
[11]			X	X		[54]		X	X	X		[89]			X	X	
[12]		X		X		[55]	X	X				[90]			X	X	
[13]	X	X	X			[56]	X	X	X		X	[91]	X	X	X	X	
[14]	X	X	X	X		[57]	X	X	X			[92]	X	X		X	
[15]	X	X	X	X		[58]	X	X	X		X	[93]	X	X	X	X	
[16]		X	X		X	[59]			X	X		[94]	X	X	X	X	
[19]	X	X	X	X		[60]	X	X	X	X		[95]		X	X	X	
[20]		X			X	[61]	X	X	X	X	X	[96]	X	X		X	
[21]	X	X		X		[62]	X	X	X	X		[97]	X	X	X	X	
[23]	X	X	X	X	X	[63]	X	X	X	X		[98]	X	X		X	
[25]	X	X	X	X	X	[64]		X	X	X		[99]			X	X	
[26]	X	X	X	X		[65]	X	X	X	X		[100]		X	X	X	
[27]		X			X	[66]		X	X	X		[101]	X	X	X	X	
[31]	X	X	X	X	X	[67]	X		X	X		[102]			X		
[35]	X	X	X	X	X	[68]	X	X	X	X		[103]	X	X	X	X	
[36]		X				[69]	X	X		X		[104]		X	X		
[37]		X	X	X	X	[72]		X	X	X	X	[105]		X	X	X	
[38]	X	X	X	X	X	[73]	X	X	X	X	X	[106]	X	X	X	X	
[39]	X	X	X	X	X	[74]	X	X	X	X		[107]	X	X			
[40]	X	X	X	X	X	[75]	X	X	X	X	X	[108]	X	X	X	X	
[41]	X	X	X	X		[76]	X	X	X	X		[109]		X	X		
[42]		X	X			[77]	X	X	X	X		[110]	X	X	X	X	
[43]	X	X	X	X		[78]		X	X	X		[111]	X	X	X	X	
[44]	X	X	X	X	X	[79]		X	X	X		[112]	X	X	X	X	
[45]		X	X	X	X	[80]	X	X	X	X		[113]	X	X	X	X	
[46]	X	X	X	X		[81]	X	X	X			[114]	X	X	X	X	
[47]	X	X	X		X	[82]	X	X	X	X		[115]		X	X	X	
[48]	X	X		X	X	[83]	X	X	X		X	[116]	X	X	X	X	
[49]	X	X	X	X		[84]	X	X	X	X		[117]	X	X		X	

B. KEY FINDINGS

This section demonstrates all the key findings we have found in the process of SLR. It is structured into five sub-sections, and each section answers one research question listed in the previous section II – A (Research Queries).

1) PROCESS ATTRIBUTES CHARACTERISTICS FOR AUTOMATION (RQ1)

Business Process Automation (BPA) is considered "*one of the digital transformation trends*" [15]. BPA's mission is eternal, "*reduce human intervention in the process*" [23], especially if combined with Artificial Intelligence [73]. BPA can also give humans the capacity to perform previously impossible activities [41, 73].

Process automation became a facilitator for all types of organizations in their strategic objectives since they will benefit from the outcomes of BPA [89]. These effects can be, for example, reducing human resources costs and increasing the accuracy and performance of the tasks performed [23, 45, 89].

This change in how organizations perform their activities can be adopted for many business areas. Can be automated processes belonging to the supply chain management [15, 23, 35, 54, 73, 75], logistics [15, 35, 73], finance [15, 23, 25, 91], banking and investment [25, 31, 73, 91], customer services [23, 25, 73, 89, 91], social support and health care services [91], operations [15, 25, 73], auditing [16, 25, 31, 44, 73], marketing and sales [23, 73], accounting and reporting [15, 23, 25, 40, 73, 89], IT/telecommunications [15, 31, 73], HR and payroll [15, 23, 54, 73], Business Process Outsourcing - BPO [31], and document management [15, 27, 40, 55, 89].

In an organization, a variety of business processes exist that work together to accomplish their goal. However, for automation, depending on the technology that the organization has, not all types of processes can be automated.

Costa et al. [61] and Moreira et al. [60], through an SLR, resume the characteristics of a process suitable for automation with RPA. This study characterizes the process by rule-based, mature, structured data, high volume, digital data, routine, few exceptions, and multiple systems. The focus of both works was RPA technology, which does not have any artificial intelligence features.

Wewerka and Reichert [31] presented the state-of-art in the area of RPA with a Systematic Mapping Study (SMS). The authors concluded that the most relevant criteria for selecting process tasks for automation with RPA are the *duration of task execution*, *complexity*, *repetitive, rule-based*, *high manual efforts*, and *digital inputs and outputs*.

The analysed papers noted a set of characteristics that must exist to automate. Table XVII summarizes the most mentioned features a process must have to be a perfect candidate for automation, the type of technology involved (if mentioned in the literature), and the respective bibliographic references.

The description of the characteristics for a process to be eligible for automation (found in the literature) is listed in the next items [14, 60, 62]:

- Rules-based: the business process must be mapped with business rules for any eventuality, without ambiguity;
- Low complexity: the process must be simple for quick bot implementation;
- Error-prone: processes that tend to make or cause errors have to be a candidate for automation;
- Very repetitive: automating tasks that are frequently repeated increases the probability of having a pleasant return on investment;
- High number of occurrences: when you automate a process that has a high number of occurrences, it helps to maximize the benefits of automation;
- Structured data: all input data in the automated process must be structured and in digital format, like a spreadsheet or database;
- Unstructured data: automation with a technology capable of working with unstructured data, like text in an email;
- Medium-high Complexity (tech with AI): we can automate a process more complex with a tool with AI features
- Normalized/structured/standardized/matured: the process must have a high state of maturation and standardization so that there are no changes in the future; the more detailed the description of the process is, the easier the configuration of the bot;
- Limited human intervention: processes that can be executed without any human intervention are ideal for automation;
- Interacts with multiple systems: a process that interacts with several systems is an ideal candidate for the technology RPA since it is performed manually, and the probability of error increases since the human has to perform access to several systems inconsistently;
- Low level of exception: the existence of exceptions can limit the automation process.

Our top five process characteristics for automation are occupied by *very repetitive*, *rule-based*, *normalized/ structured/standardized/matured, low complexity (tech without AI)*, and *high number of occurrences*, as represented in Figure 19.

TABLE XVII
PROCESS ATTRIBUTES TO BE ELIGIBLE FOR AUTOMATION

Process attribute	References
Error-prone	[14-15, 25, 31, 40, 46, 48, 55, 69, 75, 84-85, 98, 108, 116]
High number of occurrences	[14, 23, 25, 31, 35, 41, 44, 46-48, 58, 60-62, 65, 69, 75, 81-82, 91, 94, 96-98, 106-107, 112, 114, 116]
Interacts with multiple systems	[10, 14, 23, 25, 31, 35, 44, 46, 48-49, 60-62, 69, 75, 84-85, 91, 94, 97, 110, 113, 116]
Limited human intervention	[14, 31, 68-69, 75, 106, 116]
Low complexity (tech without AI)	[10, 13-15, 19, 25, 31, 35, 38, 41, 44, 46-49, 55, 60, 62, 68-69, 75-76, 85, 94, 96, 103, 106, 110, 116, 117]
Low level of exception	[14, 31, 38, 44, 61, 69, 75, 85, 116]
Medium-high Complexity (tech with AI)	[13- 14, 19, 41, 47, 49, 85, 106, 116]
Normalized/structured/standardized/matured	[13-15, 25, 31, 35, 38-39, 41, 43-44, 46-49, 55, 57, 60-63, 65, 69, 75, 77, 80, 82-85, 93, 96-98, 103, 106-107, 110-112, 116]

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Rule-based	[13-15, 23, 25-26, 31, 39, 41, 46, 48-49, 57-58, 60-63, 65, 68-69, 74-77, 82-85, 92, 94, 96, 101, 103, 106, 108, 110-114, 116]
Structured data	[14, 25, 31, 35, 38-39, 43-44, 46, 48-49, 60-61, 63, 69, 75, 84-85, 91, 106, 112, 114]
Unstructured data (tech with AI)	[13, 35, 49, 85, 91, 106]
Very repetitive	[10, 13-15, 19, 21, 23, 25, 31, 35, 39-41, 43-44, 46-49, 55, 57-58, 60-63, 65, 67-69, 73-77, 80-82, 84-85, 91-94, 97-98, 101, 103, 106-108, 110, 112-114, 116-117]

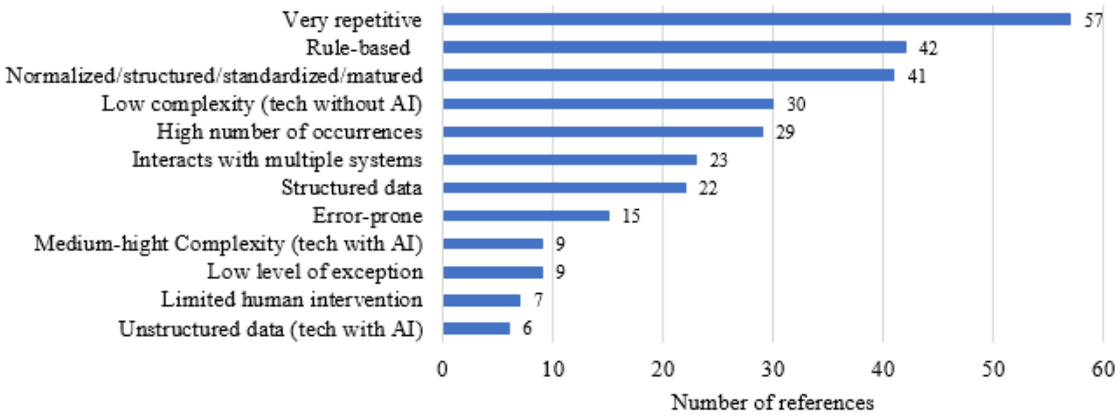

FIGURE 19. Process characteristics for automation.

A process is a candidate for automation without concern about the type of technology that the organization adopts if it is rule-based, repetitive, has a high number of occurrences, is structured, mature, or standardized, has a low level of exception, and interacts with multiple systems. Suppose the automation technology has Artificial Intelligence functionalities; the organization can automate more complex activities, semi or unstructured activities, or processes that do not yet have structured data in their input, like the text of an email or image.

How to choose which process to automate? Šperka and Halaška [14] propose a PPAFR assessment framework to "*assess the suitability of processes activities for RPA implementation with a focus on the overall performance of the process*". This framework crosses RPA characteristics, like *frequency-significant activities*, *error-prone and reworks*, *involvement of manual work*, *productivity and efficiency*, and *process flow and logic* with monitored parameters that have value in each scope. For example, Šperka and Halaška [14] call our attention to the number of reworks and the processing and waiting times for *error-prone reworks.*

Wellmann et al. [84] proposed a framework to *evaluate the viability of applying RPA in business process activities*. This framework consists of 13 criteria grouped into five perspectives, which offer different analysis aspects. The authors demonstrated and evaluated the framework by applying it to a real-life case. The five perspectives are (1) *task*, (2) *time*, (3) *data*, (4) *system*; and (5) *human*. They grouped the criteria into (1) standardization, maturity, determinism, and failure rate. In (2) perspective, frequency, duration, and urgency exist. The structuredness is the only criteria in (3) perspective. System perspective (4) concerns interfaces, stability, and the number of systems. The last perspective, human (5), grouped resources and proneness to human error criteria.

Eulerich et al. [46] developed and validated "*a three-step evaluation framework to assist auditors as they decide what activities to automate*". The first step is to answer two questions for each task that is considering automating: "*Are you allowed to use automation for the task under consideration?*" and "*If the bot is working with data, is there sufficient data/information for the bot to achieve its intended objective?*". If the answer is "no" to any question, the task is inappropriate for RPA automation, and the evaluation stops here. Step 2 consists of classifying each process based on technical feasibility (1 lower feasibility to 5 higher feasibility), considering various characteristics, and about the benefits of the bot (1 – less benefits to 5 – more benefits), considering various related items. The last step, 3, is where we plot the scores for each activity on the matrix (*feasibility* vs. *benefits*). Eulerich et al. [46] leave recommendations about where to start automation: *bots that are in Quadrant 2* [technical feasibility high and benefits of bots high] *should be prioritized for immediate development;* after their developed *Quadrant 1* [technical feasibility high and benefits of bots low] *bots should be developed. Quadrant 4* [technical feasibility low and benefits of bots high] *bots should be developed only once technical feasibility improves. Bots in Quadrant 3* [technical feasibility low and benefits of bots low] *should not be developed.*

Hofmann et al. [49] call our attention to the differences in the types of processes to automate using technologies with and without artificial intelligence. The study mentioned that "*…implementations of artificial intelligence (AI) from RPA providers… see RPA's potential among others in additional modules capable of dealing with unstructured data*" and "*Through AI technologies (e.g., machine learning), future software robots may no longer be rule-based, allowing them to self-reconfigure and to construct new software robots based on the experience of already constructed ones.*"

In the same effort to divide the type of business processes that need Artificial Intelligence support for automation from the others, Szelągowski [13] mentioned that a business process can be *static* (structured, repetitive), *structured with ad hoc exceptions*, *unstructured with pre-defined fragments*, and *unstructured*. In this work, Szelągowski [13] presents the Business Process Nature Assessment Matrix (BPNAM) that characterizes the processes by A - *Low Unpredictability and Low Knowledge intensity*; B - *Low Unpredictability and High Knowledge intensity*; C - *High Unpredictability and Low Knowledge intensity*; and D - *High Unpredictability and High Knowledge intensity*. In this BPNAM, the processes classified as belonging to quadrants A and B do not need artificial intelligence (also considered machine learning) because they are predictable. However, with their high unpredictability, quadrants C and D will need Artificial Intelligence and Machine Learning mechanisms to make their automation. This process in small and medium-sized enterprises should not last more than 2 to 3 days [13]. He has also mentioned that most business processes have low unpredictability, so using RPA tools for their automation is sufficient.

Martinek-Jaguszewska and Rogowski [41] expose a ranking of types of processes in terms of "*automation capabilities based on the increasing complexity and novelty*" as follows: (1) *current simple processes*; (2) *standard, routine, and repetitive tasks*; (3) *complex but standard processes*; (4) *nonstandard processes*; and (5) *completely new, complex processes*. The higher the ranking, the more intelligent the automation tools have to be.

Closing this topic remains to mention the notation most used to describe a process that can be the target of automation. This description process becomes especially important in the organization since the administrators can read the process, the operational resources, the IT professional, and the technology itself through a graphic representation. Therefore, in our review of articles, we found, for the representation of the business process, the following notations and languages: Business Process Model and Notation (BPMN) [14, 16, 27, 40, 55-57], Event-Driven Process Chain (EPC) [16], Integrated Definition Language (IDEF) [16].

Neto et al. [56] and Nurhayati and Fitrisari [27] mentioned that BPMN is the leading technology (standard model) for modeling business processes. BPMN was created by the Business Process Management Initiative (BPMI) to provide a notation that could be interpreted and manipulated by all business intervenients (technical and non-technical personnel) to understand a process [27, 56]. BPMN is an ISO and OMG standard [56]. BPMN diagrams can be executable and interpreted by process engines that can automate the process [56], especially two elements: events (start, indirect) and tasks (user, subprocess) [55].

2) TYPES OF TECHNOLOGIES/TOOLS SUPPORTING BPA (RQ2)

The authors think that Business Process Automation (BPA) can be achieved with a large type of technology/tools, which can begin with a simple macro from Excel and culminate with Blockchain, for example. An organization is a case of independent analysis. Therefore, the most suitable technology for one may not be suitable for another. Technology can be seen as an enabler that makes enterprises more and more competitive [10]. In the SLR, we found references to simple and more complex technology. Table XVIII summarizes the most mentioned types of technology/tools used in business process automation, with the respective bibliographic references.

Our top three for process automation are Robotic Process Automation (RPA), Intelligent or Cognitive Process Automation (IPA or CPA), and Workflow Manager/Business Process Management System (WfMS/BPMS), as represented in Figure 20.

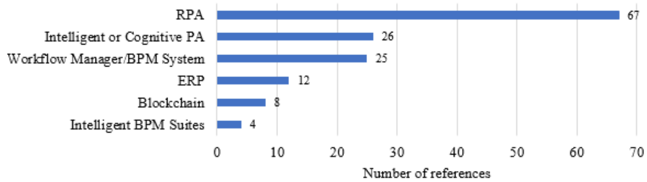

FIGURE 20. Types of technology for process automation.

Workflow Management Systems (WfMS) are tools that define models and rules for designing business processes and models for the respective execution and workflow analysis [12]. The outcomes of the WfMS are provided by all business process rules predefined by human resources and follow if-then-else conditions [42]. The target business process for automation with WfMS is non-cognitive knowledge and service work [42]. Business Process Management Systems (BPMS), through delegation of the execution of tasks to software with APIs, allow BPA [38]. Seymour and Koopman [109] and van der Aalst et al. [115] mentioned that BPM was an evolution of Workflow Management, so the systems that support that discipline, BPMS, were the evolution of WfMS. Flechsig et al. [57] defined BPMS as a system that has tools for *modeling* (process modeller), *analysis*, and *execution* (process engine) of the business process. These systems have features for *process orchestration, resource management, process monitoring, and process analysis* [38]. Alves et al. [16] define BPMS as a suite of tools supporting business processes' modeling, analysis, improvement, and automation. BPMS is considered beneficial to small and medium enterprises in the area of automation [72].

Enterprise Resource Planning (ERP) is *very popular* in organizations because "*it offers value through automation and integrating internal business processes, helping to reduce cycle time"* [52]. So, this technology can be viewed as a means of business process automation [52-54] to "*increase productivity, reduce costs, and meet customer requirements"* [54]. Seymour and Koopman [109] mentioned that ERP was the evolution of WfMS/BPMS.

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ERP systems digitalize the flow of information and provide functionalities to support operational business processes [53]. ERP "*can maintain and automate standardized work while ensuring that work is done in exact sequence and format*" [53]. Aboumasoudi et al. [100] define ERP as a system that connects accounting, inventory control, financial operations, supply chain, and human resources functions. From a process-oriented perspective, ERP supports the organization and sets standards for its business process [100]. ERP systems can lead to increased efficiency in business processes [100].

Robotic Process Automation (RPA) is considered a *first step in digital transformation* [15, 25, 35, 73], *recent technology* [15, 31, 38, 47, 58, 61, 73, 108] and is considered in the *growing adoption process by organizations* [14-15, 31, 37, 46, 49, 61, 68, 73, 75, 79, 91]. In his study, Sobczak [73] found that "*large and very large enterprises were first to commence deploying robotic process automation solutions*." However, "*small and medium-sized enterprises are just beginning to implement this approach*." RPA is consumed by all kinds of organizations, from small to very large ones [15, 73].

RPA is an automation technology for processes with *repetitive* [10, 14-15, 19, 25, 31, 35, 39-46, 60-62, 68, 75, 84, 91, 93, 98, 108, 116], *routine* [14-15, 19, 23, 26, 31, 35, 42, 44, 46, 58, 60-61, 91, 98], *simple rules or rule-based* [14-15, 25-26, 31, 35, 39, 42, 44, 49, 58, 60-62, 68, 75], with *data inputs and outputs structured* and readable by a machine [14, 31, 39, 41-42, 45, 61-62, 73, 75, 84, 98] tasks.

This technology is characterized as a *software* robot (*bot*) [10, 15, 19, 31, 39-40, 42-49, 57-58, 73, 79, 91, 116] that *interacts with other systems mimicking humans* [10, 14, 19, 23, 25, 31, 35, 38-40, 42-45, 47-49, 57-58, 61, 68, 73, 75, 79, 84, 91, 93, 98, 101, 108], does *not require classic coding* [14, 15, 42, 48, 57, 73, 91], and *independent of the others systems* that interact [14, 19, 31, 39, 42, 44, 47-49, 57, 62, 75, 91].

RPA tools benefits are essentially *reducing costs* [15, 37-38, 43-44, 60-62, 68, 75, 84, 91, 98, 101 116,], *improving employee experience freeing employers from routine tasks* [14-15, 37, 42, 61, 84, 91, 116], *reducing error ratio* [15, 37-38, 44, 60-61, 68, 91, 98, 101], *time-saving* [15, 37-38, 43-44, 60, 68, 75, 84, 98], and *high availability* [15, 44, 60, 116].

Sobcza [73] refers to Robotic Desktop Automation (RDA) and Cognitive Robotic Process Automation (C-RPA) or Intelligence Robotic Process Automation (I-RPA) as tools related to RPA. Authors mention that exists two types of RPA: (1) bots only for performing repetitive and highly frequent and structured tasks based on simple and clear rules (RPA and RDA); (2) bots that were trained and perform complex and unstructured tasks, while being flexible and adaptive, due to combination with Artificial Intelligence (AI) features (C-RPA or I-RPA) [14, 42, 49, 73].

TABLE XVIII
TYPES OF TECHNOLOGIES/TOOLS FOR PROCESS AUTOMATION

Types of Technology	References
Blockchain	[8, 10, 15, 19-20, 35-36, 77]
Enterprise Resource Planning	[21, 51-54, 66, 86, 100, 104-105, 109, 115]
Intelligent Business Process Management Suites	[13, 21, 36, 41]
Intelligent or Cognitive Process Automation	[10, 14-15, 19, 23, 26, 31, 35, 41-42, 44-45, 49-50, 61, 63, 73, 75, 81, 85, 91, 93, 95, 106, 110, 114]
Robotic Process Automation	[10, 13-15, 19, 23, 25-26, 31, 35, 37-50, 57-58, 60-65, 68-69, 73-85, 91-98, 101, 103, 106-108, 110-112-114, 116-117]
Workflow Manager/Business Process Management System	[12-13, 16, 21, 27, 31, 36, 38, 40-42, 51, 55-57, 69, 72-73, 79, 85, 91, 93, 107, 109, 115]

Intelligent Process Automation (IPA) was proposed to take advantage of AI in the limitations of RPA, which refers to combining AI, cognitive automation, deep learning, and machine learning with RPA [44]. IPA learns "*how people make decisions and may be able to perform complex tasks faster and better"* [44]. Herm et al. [47] agree with the last quote. Considering AI features in process automation will have strong effects on human labour, in the process itself, IS ecosystems, and customer experiences, organizations must have a strategic approach for its implementation [49].

Organizations must collaborate to achieve competitive and greater goals [36]. This environment requires extensive information exchange, and the design and management of the inter-organizational business process is difficult [36]. Blockchain is a recent technology that can bring innovation in how these organizations interact [10, 36]. With this kind of technology, inter-organization business processes can be supported, overcoming insecurity issues [10, 19-20, 35], increasing trust, and reducing fraud [19-20, 35]. This technology can also be used for process monitoring and encryption [19].

Recovering the top of the technologies/tools, the authors of this SLR believe it is important to compare RPA and I/CPA and RPA and BPMS. This comparison was inspired by Wewerka and Reichert [31] and then completed by other authors. So, beginning with RPA and I/CPA, their principal differences are summarized in Table XIX.

TABLE XIX
RPA VERSUS I/CPA

Criteria	RPA	I/CPA	References
Standardization	High	Low	[14, 31, 42, 49, 63, 73, 85, 106]
Type of data	Structured	Unstructured	[14, 31, 42, 49, 63, 73, 85, 110]

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Criteria	RPA	I/CPA	References
Type of decisions	Rule-based	Knowledge/ experience-based	[14, 31, 42, 49, 63, 73, 85, 106, 110]
Outcome	Deterministic	Probabilistic	[31, 63, 85]
Exceptions	Demand human intervention	Trigger machine learning	[31]

The principal differences between RPA and I/CPA were reported; the next goal is to know how to differentiate RPA and BPMS solutions. Gabryelczyk et al. [21] mentioned that the technology has evolved from workflow systems/BPMS and ERP to RPA, artificial intelligence, and more. Sobczak [73] and Sobczak and Ziora [91] assumed that RPA and other classic solutions for business automation have common objectives: increase efficiency and minimize costs while ensuring the quality of products or services delivered by the automated process. However, these objectives are achieved using different means [73].

Wewerka and Reichert [31] state that "*understanding these differences is fundamental for appropriate RPA and BPM methods in an enterprise context.*" The authors concluded that tasks with *low complexity* and *high frequency* must be the best for BPM software, tasks with *medium complexity* and *medium/high frequency* are suitable for RPA, and the last ones, with *low frequency* and *low complexity,* can be only performed by humans. RPA and BPMS are two types of technology that combine to form a winning team. They complement each other [31, 38]. In Table XX, we present some of these differences in the literature.

From the literature, we can say that the perfect solution to automate a process is not always the same. The authors found RPA somehow in various automation examples. Gabryelczyk et al. [21] mentioned using RPA to reinforce process improvement and the automation of processes using an ERP system. Costa et al. [61] also cited ERP with RPA. Flechsig et al. [75] state the use of RPA for full automation of long-tail processes through ERP and e-procurement systems. Šperka and Halaška [14] exemplify the use of RPA to transfer data from CRM or ERP systems to another application. Hartley and Sawaya [35] demonstrated the use of RPA with ERP and machine learning in process automation in their study. Flechsig et al. [57] persuaded an integrated platform for BPMS and RPA.

3) INFLUENCING FACTORS FOR BPA IN SMES (RQ3)

Information Technology (IT) is one pillar of competitiveness in SMEs (Small and Medium-sized Enterprises) since it can improve the speed, accuracy, and efficiency of the processes [90]. Productivity can be increased by 50% using IT [90]. With RPA, organizations can increase productivity by 86 % and quality by 90 %, while office costs should be reduced by 59 % [61]. SMEs have become clients of automation and are trying to implement this work paradigm in their processes [15, 90]; however, about 40% of RPA projects fail to reach the general goal of decreasing implementation time and implementation costs, among other costs [23]. 30 to 50% of RPA initiatives fail [15].

TABLE XX
RPA VERSUS BPMS

Criteria	Rpa	BPMS	References
Objective	Automation of tasks by a *bot* software	Automation of tasks involving the interactions at running time	[14, 31, 38, 85]
Focus	Change "where" work is done	Change "how" work is done	[14, 31, 69, 73, 79, 85, 91]
Type of technology	Non-invasive, lightweight IT	Invasive, heavyweight IT	[14, 31, 48, 57, 68, 73, 75, 79, 85, 117]
Risk of failure	Low	High	[73, 79]
Implementations Costs	Low	High	[31, 48, 57, 69, 73, 117]
Time for implementation	Low	High	[48, 57, 69, 73, 117]
Programming skills	No	Yes	[14, 48, 57, 68-69, 85]
Problems	Privacy, security issues	High complexity	[31]

Herm et al. [47] mentioned that "*although RPA is … considered a straight-for-forward to implement… export knowledge is necessary to create reliable and scalable software robots… As a result, up to 50% of … RPA implementations are estimated to fail*". Eulerich et al. [46] estimated that "*40% of respondents struggle to identify and prioritize potential RPA initiatives, which is the number one problem they find for adopting RPA … as evidence, many of their bot implementations fail*".

The analyzed literature revealed the factors shown in Table XXI that can negatively influence BPA. For automation, we have considered using many types of automation or improvement methods to reduce costs and increase productivity and efficiency.

Several factors can influence the success of an automation project. Tarutė et al. [102] mentioned internal and external factors that can influence digital transformation in SMEs (Small and Medium-sized Enterprises). The *internal* group considered capabilities fit, resource fit, and changes in the business model. The *external* group factors considered external capabilities and resource fit, governmental regulation, and industry-related factors. Flechsig et al. [75] mentioned in their multiple case studies of implementing RPA in purchasing and supply management that *digital readiness* results from the evaluation of technical, organizational, and environmental factors. They used a scale with four levels: low, low-medium, medium, and high. The authors concluded that advanced RPA users have high digital readiness. Digital readiness can be a concern for all automation candidates.

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Martinek-Jaguszewska and Rogowski [41] mentioned that automation's benefits depend on factors like "*automating the right processes with the right technologies, adjusting the organizational structure, and ensuring all actions align with the business strategy.*"

Our top five problems for successful adoption of automation are occupied with the negative factors: *not having a methodology for automation*; *lack of knowledge about technology/automation; cost of the solution; inertia and resistance of employees*; and *lack of solution governance*, as represented in Figure 21. Feio and dos Santos [45] leave a piece of advice for organizations to adopt before automation: "*... must develop a roadmap and strategy, ensure technology readiness, guarantee human resource capabilities implement change management, maximize the impact through full use of IPA solutions portfolio and have the mindset of building rapid minimum viable products*". In the same path, we found Lazareva et al. [23] that mentioned "*… that out of 400 companies, 63% were not able to implement process automation projects on time. The main reason is … the lack of an action plan and guidelines…*". In SMEs, it is very unlikely that this procedure can be done without any external consult help [72].

Stravinskienė and Serafinas [15] classified the automation pitfalls into five categories: organizational, process, implementation, technical, and post-implementation. We divide these challenges into the *organizational, social, human, technological, implementation process*, and *governance aspects*. These can group the negative automation factors like:

- organizational aspects: lack of budget; poor management/overcentralized decision-making/rigid hierarchical structure; lack of transparency; low flexibility/agility; lack of skilled human resources; lack of strategic alignment (business goals and IT);
- social and human aspects: inertia and resistance of employees; ethical and moral issues;
- technological aspects: lack of innovation; lack of digitalization; lack of knowledge about technology/automation;
- implementation process aspects: concerns in cybersecurity and data privacy; not having a methodology for automation; lack of knowledge about organizational processes;
- governance aspects: lack of solution governance.

Despite the organizational aspects of the group having more pitfalls, we can conclude that at least one of our top five problems belongs to each group. This revealed that the procedure of business process automation involves the entire organization and cannot be analyzed from one perspective. Automation involves organizational, human, technological, implementation, and governance aspects.

Some authors have analyzed the criteria for automation success from a positive perspective, not a negative one. Chatzoglou et al. [105], Reitsma and Hilletofth [86], and Kouriati et al. [66] mentioned a set of Critical Success Factors (CSF) to implement ERP. Therefore, to obtain all benefits from BPA or minimize the negative factors of automation explained before, the organization has to orchestrate complementary BPA approaches, like WfMS, RPA, and other BPA technology with AI or ML features [42]. However, organizations still face a high level of doubt in choosing which technologies to adopt; "*knowledge about digital technologies is vital*" [8]. Some authors recognized that implementing automation is essential in some BPM initiatives [15-16, 23, 72]. All the factors found are listed in Table XXII.

Another important positive factor is *communication*. With this powerful weapon, organization management can inform human resources about the changing routines of work, trying to dissipate their concerns about losing jobs and how to perform new tasks, among other items [15, 40, 43, 75, 104]. With these communication plans, organizations can minimize the *inertia and resistance of employees*, for example.

Adopting *low-cost* automation solutions (for example, RPA) minimized the problem of *lack of budget* [31, 35, 38, 47, 49, 91]. RPA can also be implemented in a *low time* compared to the traditional ways so that organizations can achieve faster automation [31, 47, 49, 91].

TABLE XXI
FACTORS THAT CAN NEGATIVELY INFLUENCE BPA

Factor with negative influence	References
Concerns in cybersecurity and data privacy	[7-8, 19, 40, 45, 61, 68, 106, 114]
Cost of the solution	[7, 11, 31, 35, 37, 40, 43, 45, 49, 54, 57, 60, 64, 66, 72, 74-75, 78-80, 86, 89-90, 95, 100, 105-106, 108, 110-111, 114-115]
Ethical and moral issues	[15, 19, 23, 31, 41, 43, 45, 62, 64-65, 68, 73-79, 81, 89, 106, 109, 112-114]
Inertia and resistance of employees	[6, 15-16, 19, 23, 31, 35, 40, 43, 45, 50, 60-61, 64-65, 73, 75, 77-79, 84, 87, 89-91, 106, 109, 112-113]
Lack of digitalization in process/organization	[23, 35, 42, 50, 57-58, 61-63, 73-75, 79, 87, 90, 93-94, 100, 102-103, 111, 114]
Lack of innovation in the organization	[7, 11, 13, 23, 31, 35, 37, 43, 56, 75, 102]
Lack of knowledge about organizational processes	[6, 11, 13, 16, 23, 40, 43-44, 50, 52, 56-57, 61, 66, 68, 72-73, 91, 103, 108-109]
Lack of knowledge about technology/automation	[7-8, 11, 13, 15, 19, 23, 31, 38-41, 44-45, 47, 50, 54, 56-58, 60-62, 64-66, 72-73, 75-79, 81-83, 87, 89-91, 95, 97, 99, 101-103, 105, 108, 110-114]
Lack of skilled human resources	[7, 11, 13, 15, 23, 31, 35, 38, 42-43, 46-47, 54, 57, 73, 89-90, 102, 106, 109, 114]

Factor with negative influence	References
Lack of solution governance	[11, 15, 23, 31, 35, 39, 41-42, 45, 47, 49, 52, 54, 57, 60, 64, 73, 75, 78, 80-81, 83, 89, 95, 109, 114]
Lack of strategic alignment (business goals and IT)	[11, 13-14, 23, 31, 35, 41-43, 45, 49, 54, 56-57, 68, 75, 100, 102, 109]
Lack of transparency of the process or change	[11, 35, 40, 45, 75, 106, 114]
Low flexibility/agility in the process	[6, 11, 15, 35, 37, 49, 88]
Not having a methodology for automation.	[8, 11, 13, 15, 23, 26, 31, 35, 38-47, 49-50, 54, 56-58, 59, 61-62, 64-67, 73, 75-84, 86, 89, 91, 93-95, 99, 103, 105, 108, 110-111, 113-114, 116]
Poor management/overcentralized decision-making/rigid hierarchical structure	[7, 11, 15, 41, 45, 49, 56, 61, 66-68, 75, 87, 90, 99, 102, 106]

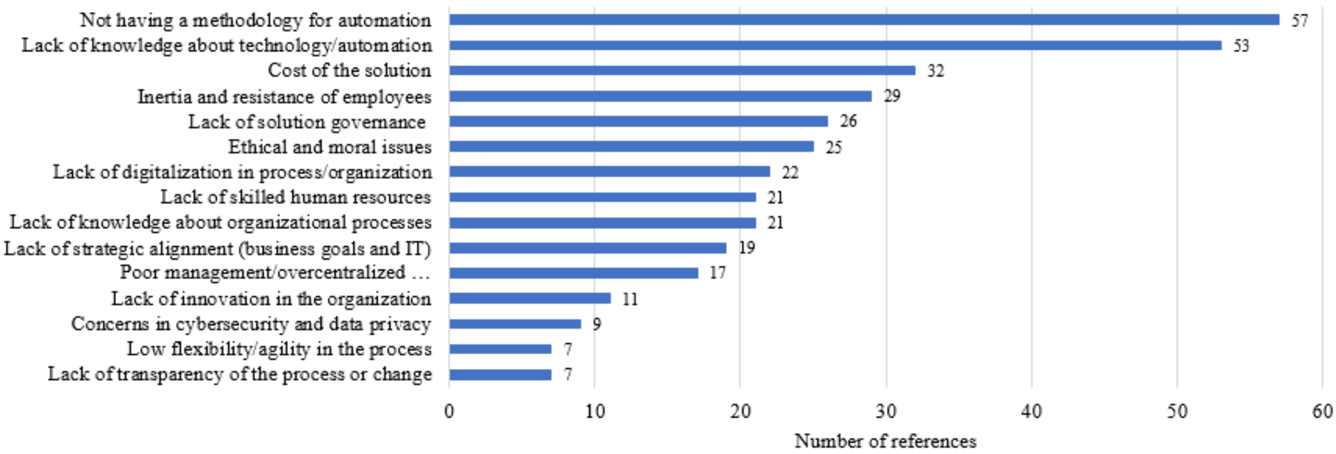

FIGURE 21. Number of references for each factor that can negatively influence BPA.

Our top five items for successful adoption of automation are occupied with the positive factors: the existence of a *plan of communication*, providing *staff straining* to deal with all the changes, the existence of a plan for *change management*, the existence of a *culture/politics/organizational structure that supports changes*, and knowledge about digital technologies, as represented in Figure 22.

As seen in the list of negative factors (Table XXI), the lack of knowledge is a handicap for automation. However, suppose the organization provides staff training in the area of automation [7, 15, 25, 35, 40, 89], in how to use the new bots, and how to deal with this new paradigm [7-8, 14-15, 25, 35, 73]. In that case, this can positively influence the automation procedure.

This type of project can be complex, as it can require many changes in the organization, both in terms of processes and in the organizational and technological infrastructure, so if do not exist *change management* [7, 15, 23, 25, 66-68, 73, 75, 86-87, 100, 104, 106] and *process management* [7, 14, 23, 73, 75], the procedure of automation of business process runs risks of not being successful. All these changes, innovations, and restructurings must always be accompanied and supported by the company's management [7, 14, 25, 35, 73, 75].

TABLE XXII
FACTORS THAT CAN POSITIVELY AFFECT BPA

Factor with positive influence	References
Change management	[7, 15, 23, 25, 66-68, 73, 75, 87, 100, 104, 106]
Communication about job disruption	[15, 40, 43, 75, 106]
Culture/Politics/Organizational Structure that supports changes	[7, 14, 25, 35, 66, 68, 73, 75, 86-87, 100, 104-105]
Existence of BPM	[15, 16, 23, 57, 66-68, 72, 86, 105]
Knowledge about digital technologies	[7-8, 14-15, 25, 35, 66, 73, 86, 104-105]
Low cost of automation solution	[31, 35, 38, 47, 49, 68, 91]
Low time of implementation	[31, 47, 49, 68, 91]
Orchestration of various kinds of BPA approaches	[35, 38, 42, 73]
Plans of communication	[7, 15, 25, 35, 40, 43, 66-68, 75, 86-87, 100, 104-105]
Process management	[7, 14, 23, 66, 73, 75, 86, 104-105]
Staff training	[7, 15, 25, 35, 40, 66, 68, 86, 89, 100, 104-106, 114]

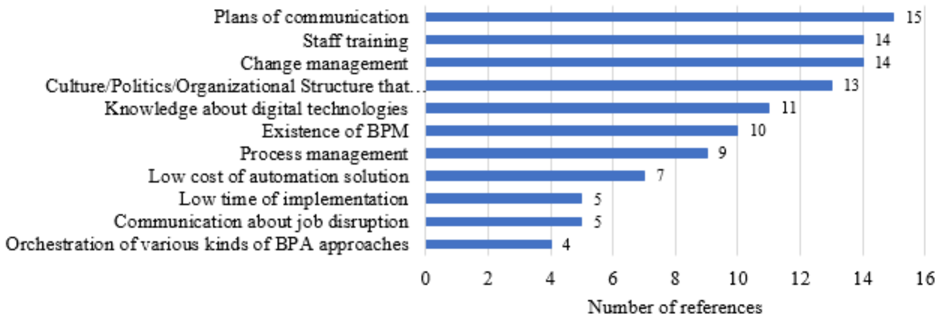

FIGURE 22. Number of references for each factor that can positively influence BPA.

4) EVALUATION OF THE SUCCESS OF BPA IN SMES (RQ4)

Once the task of choosing the process to automate and the supporting technology is finished, and the environment is prepared for the transition of work paradigm, it is important to reflect on how the organization can perceive the success or advantages of the new scenario compared to the previous one [53]. In van der Aalst et al. [115], it can be found that to measure process improvement, organizations can use a lot of *Key Performance Indicators* (KPIs), also known as *Process Performance Measures* (PPM). In the literature reviewed, the evaluation was mostly related to the benefits of automation (see Table XXIII). The organizational competitive advantage came from measuring the performance of the business process [88].

Van der Aalst et al. [115] have grouped some KPIs or PPM in dimensions: *time*, *quality*, and *cost*, among others. For measuring time, the organization can use cycle time or waiting time (non-value adding time); for cost, the authors mentioned cost per execution, resource utilization, and waste; and for measurement, the quality of customer satisfaction, errors rate, and other violations. Šperka and Halaška [14] and Syed et al. [62] summarize the automation benefits in four groups: *operational efficiency*, *quality of service*, *implementation and integration*, and *risk management and compliance*. Wewerka and Reichert [31] also clustered into four categories (excluding the human aspects): *speed*, *availability*, *compliance*, and *quality*. Liutkevičienė et al. [53] set as typical metrics measurements of *productivity*, *quality*, *time*, and *customer satisfaction* for measuring the performance of a business process.

Stravinskienė and Serafinas [15] and Hofmann et al. [49] defined two groups of KPIs for evaluating RPA impact on business: *internal factors* that include job satisfaction, process acceleration, cost savings, and employee productivity enhancement, and *external factors* including customer satisfaction, cooperation with other organizations. In the same area, Hofmann et al. [49] mentioned that *internal factors* can mediate external performance. The authors also mentioned that the definition of KPIs for an adequate evaluation in this area is still lacking.

Gabryelczyk et al. [21] mentioned three core benefits of using BPM: efficiency, effectiveness, and agility. Šperka and Halaška [14] concluded in their study that there is no *proposal of process performance indicators (PPIs)* that we can use to assess the benefits of the overall performance of automated processes. These authors proposed a PPAFR assessment framework with many monitored parameters. For example, to verify *productivity and efficiency*, Šperka and Halaška [14] enumerated the *average processing times of activities*, *average waiting times of activities*, *total processing times of activities*, *total waiting times of activities*, *potential time savings*, and *average cycle time*.

Our top five benefits in the successful adoption of automation are occupied with *reducing process time* (with 68 references), *increased effectiveness* (with 60 references), *reduced cost* (with 54 references), increased *accuracy* (with 33 references), and *process improvement* (with 33 references) as represented in Figure 23. We believe these five factors can be transformed in KPIs in most general organizations.

So, a range of KPIs and performance measurement systems exist to evaluate the improvements created by adopting automation. Each organization is unique and must define its metrics [53]. The choice of KPI should reflect the organization's strategic objectives [115].

TABLE XXIII
BENEFITS OF AUTOMATION THAT CAN BE FORMULATED FOR MEASUREMENT OF THE SUCCESS OF BPA

Benefits of BPA	References
Employee engagement or satisfaction	[14-15, 19, 37, 43, 64-65, 73, 75-76, 78, 82, 84-85, 91, 94-96, 111, 113, 116]
Improving customer satisfaction	[11-12, 14, 19, 25, 31, 35, 37, 53-54, 59-61, 65, 73, 75, 87, 91, 98, 105, 108]
Increased accuracy	[31, 62-65, 67-69, 74-76, 78-80, 82, 84, 87, 92-99, 101, 103, 105, 108, 110-116]
Increased availability	[31, 65, 69, 73, 76, 80, 92-95, 97, 101, 111, 113, 116]
Increasing effectiveness	[11-12, 14-15, 19, 21, 25, 31, 35, 38, 40-41, 43, 45-46, 49, 53-54, 60-69, 72-80, 82, 85-87, 89-97, 100-101, 103, 106, 108, 110-114]
Mechanisms for auditing	[12, 15, 19, 23, 25, 31, 39-40, 49, 62-63, 72, 74, 76, 78, 80, 87, 92, 94, 96-97, 99, 103, 110-113]

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Benefits of BPA	References
Process improvement	[15, 21, 45, 49, 53, 60, 62, 65-66, 69, 73-75, 80, 84-87, 92, 95-96, 98-100, 103, 105, 108, 111-116]
Reducing costs	[10, 14-15, 19, 25-26, 37, 39, 41, 43-46, 48-49, 54, 59-62, 65-69, 72-76, 79-82, 84, 87, 89, 91-92, 94-99, 101, 103, 105-106, 110-114, 116-117]
Reducing process time	[10-12, 14-15, 19, 21, 23, 25, 31, 35, 37-39, 41, 43-46, 48-49, 52-54, 59-65, 67-69, 72, 74-80, 82, 84-85, 87, 89, 92-101, 103, 105-106, 108, 110-116]
Resources for difficult/intelligent tasks	[14-15, 19, 31, 35, 37, 43, 49, 61, 69, 73, 75, 85, 91, 108, 116]

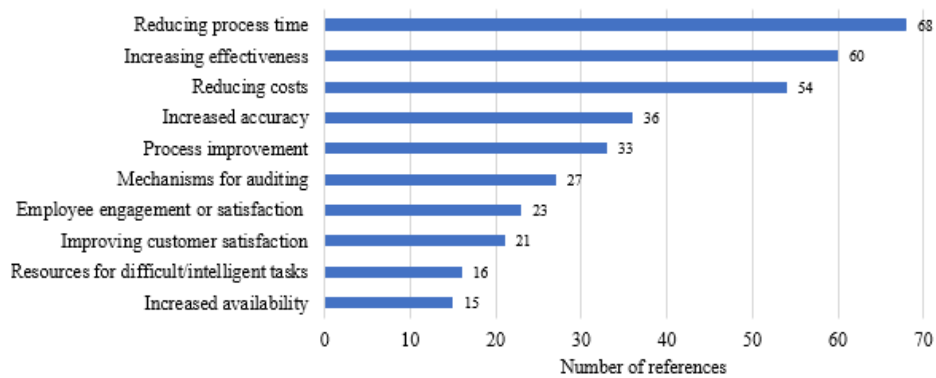

FIGURE 23. Number of references for each benefit of automation that can be formulated for measurement of the success of BPA.

5) FRAMEWORKS/ROADMAPS/GUIDELINES/METHODOLOGIES FOR BPA IN SMES (RQ5)

As we have already stated in this study, one of the most cited problems in the automation business process is the lack of a methodology for automation (see Figure 21). So, this section lists, in chronological order, all the roadmaps, guidelines, methodologies, or frameworks for automation found in the literature review (Table XXIV). Also, the findings have considered the technologies of Robotic Process Automation (RPA), Enterprise Resource Planning (ERP), Workflow Management Systems (WfMS), and Business Process Management (BPM) (technologies listed in Table XXI). This SLR found 28 models to support the procedure of business process automation in organizations. About 39% are frameworks [20, 23, 37, 44-45, 47, 50, 53, 56, 58, 88], 29% are guidelines [27, 39-40, 48, 61, 72-73, 83], 18% are roadmaps [6, 25, 35, 51, 75], and 14% are methodologies [8, 16, 31, 38] (Figure 24).

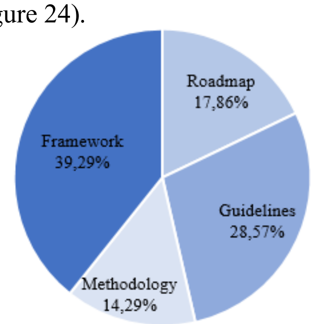

FIGURE 24. % of the types of models found in the SLR.

Most models belong to the year equal or superior to 2019, 21 of 28 works (Figure 25). Coincident with the time of the world pandemic COVID-19 and the consequent necessities of the organizations to perform their business, perhaps.

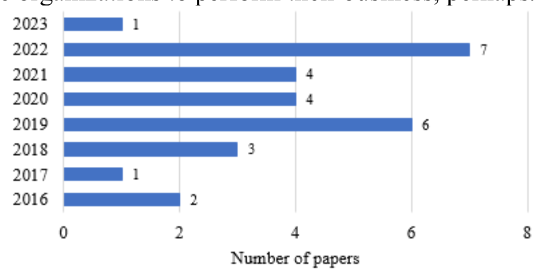

FIGURE 25. Number of papers with models for BPA per year of publication.

Also, this SLR noticed a possible gap in studies about the application of models to guide the overall procedure of business process automation, specifically in SMEs. In the Organization column from Table XXIV, it is possible to see that SMEs are not asked in most studies and are only explicitly referred to in six papers [47, 53, 58, 72-73, 83]. Another evidence from this SLR is the large range of sectors of activities of the organizations that can adopt automation (recall the answer to the RQ1). In the column Sector of Table XXIV, the area of business can be assessed.

In these 28 studies, there is also a focus on a wide spectrum of types of technology (see column *Technology/Tool* of Table XXIV). For example, there is the simple workflow system [27, 40, 51, 56], more complex intelligent process automation [35, 37, 45, 50], and lastly, cases of integration with various types of technologies [6, 35, 38].

Almost every 28 studies have a background in literature. Some use the interview method to test requirements for successful automation or validate the model. The authors of this SLR also welcomed the use of the case study method. With these case studies, there is empirical evidence that the benefits proclaimed as coming from automation do or do not happen. All methodologies employed can be consulted in the column *Methodology* from Table XXIV.

From all the 28 listed references, their stages, guidelines, and topics of interest have been analyzed. The authors of this SLR think that a procedure for implementing automation in SMEs must include the aspects marked in Table XXV: roles in procedure; methodology for the procedure; the approach for the project; study process to automate; technology/tool for automation; integration of the technology information systems; development of the automation; test of automation; assessment of the automation; deployment of the process; monitoring the process; change management and upscaling of the automation. Please note the Description (column in Table XXV) for more details on each aspect considered important in the automation procedure.

TABLE XXIV

LIST OF GUIDELINES, ROADMAPS, METHODOLOGY, OR FRAMEWORKS FOR AUTOMATION (CHRONOLOGICAL ORDER)

Paper	Type	Sector	Technology/Tool	Methodology	Organization
[50]	Framework	Various areas	Cognitive BPM	Literature	–
[27]	Guidelines	Health Insurance	Workflow System	Case Study	Public Sector
[6]	Roadmap	Public sector	BPM/BPR/BPA	Case study	Public Sector
[16]	Methodology	Court of Accounts	BPM	Literature/Case Study	Public Sector
[8]	Methodology	Production, Services, and Consulting	BPM Process Improvement	Literature/Action Design Research	From 40 to 6200 employees (medium to large enterprises)
[37]	Framework	Areas of insurance	IPA/IRPA	Literature	–
[20]	Framework	–	BPM	Literature	–
[72]	Guidelines	Optical retail business	BPM	Literature/Case Study	SMEs
[44]	Framework	Auditing area (accounting)	RPA	Literature/Case Study	–
[35]	Roadmap	Supply chain	RPA/AI/ML/BCK	Interviews	Large, mature
[40]	Guidelines	Railway company	Workflow System	Case Study	Public Sector (>320,000 employees)
[83]	Guidelines	-	RPA	Literature	SMEs
[48]	Guidelines	-	RPA	Literature (SMS)	-
[38]	Methodology	Financial services on application	Integration RPA in BPM	Literature/Case Study	–
[88]	Framework	–	–	Literature	–
[39]	Guidelines	Banks, mail organizations, and the public sector	RPA	Literature/Delphi Study/Case Studies	Public sector and others
[75]	Roadmap	Supply chain	RPA	Interviews/Case Studies	Public and private company
[73]	Guidelines	–	RPA	Surveys/Literature	Polish enterprises from <50 persons (FTE) to >=5000 persons (FTE)
[25]	Roadmap	Financial	RPA	Case Study/Interviews/Literature	Private bank
[31]	Methodology	–	RPA	Literature	–
[61]	Guidelines	–	RPA	Literature	–
[45]	Framework	–	IPA/RPA	Literature/ Design Science Research/Interviews/Case Study	–
[47]	Framework	–	RPA	Literature/ Design Science Research	Including SMEs
[23]	Framework	Retail sector	RPA	Literature	–
[53]	Framework	SMEs and/or service companies	ERP	Literature/Case Study	SMEs
[56]	Framework	Logistics and seaport domains.	Workflow System	Literature/Artefact/Case Study	–
[51]	Roadmap	Pneumatic and electrical automation technologies	Workflow System	Cases studies	Large enterprise
[58]	Framework	-	-	Literature/ Design Science Research	SMEs

The procedure of business process automation can be divided into three moments: pre-implementation, implementation, and post-implementation. For each phase, some activities and requirements must be performed. Besides these three stages, it is important to have present in all-time plans for change management, communication, and governance. It is important also to define the methodology, roles, and approach for the procedure. Revisiting Table XXV, the items presented can be divided into implementation procedure stages, as noted in Table XXVI.

TABLE XXV
LIST OF ASPECTS THAT WERE CONSIDERED IMPORTANT FOR AUTOMATION

Item	Symbol	Description
Roles in procedure	ROLL	who or roles to perform the procedure; external or internal elements to the organization;
Methodology for procedure	MET	automation of all processes; one by one; pilot; proof-of-concept (PoC)
Approach for project	APPR	the approach for implementation, for example, agile, lean, or other
Study process to automate	PROC	choose and document the as-is as the to-be process after implementation
Technology/tool for automation	TECH	develop or choose which type of technology/tool is more suitable for the process to automate
Integration TIS	INTG	issues about how to integrate automation into the actual information systems infrastructure
Development of automation	DEV	development of the process automation (bot, workflow, …)
Test of automation	TEST	test of the automation process
Assessment of the automation	ASSE	measurement and evaluation of this automation, comparison of the before and after automation
Deployment of the process	DEPL	deployment of the process automated
Monitoring the process	MONI	monitoring the process, verifying its behavior, error-rates, performing maintenance
Change management	CHGM	the existence of a protocol for change management
Upscaling of automation	UPSA	politics of upscaling

TABLE XXVI
ITEMS GROUPED IN STAGES OF THE PROCEDURE OF BUSINESS PROCESS AUTOMATION

Stage	Item
During all procedure	Roles in the procedure (ROLL); Methodology for the procedure (MET); Approach for the project (APPR); Change management (CHGM);
Pre-implementation	Study process to automate (PROC);
Implementation	Technology/tool for automation (TECH); Assessment of the automation (ASSE); Integration TIS (INTG); Development of automation (DEV); Test of automation (TEST); Deployment of the process (DEPL);
Post-implementation	Monitoring the process (MONI); Upscaling of automation (UPSA);

To compare the steps cited by each author, we present the works of the authors listed in order by year of publication in the next Table XXVII. The bottom positions of our ranking of models (considering the total of aspects that each fulfils), with a total of 3, are occupied by [8], [35], [73], and [83]. [8] and [35] considered the study of the process to automate. Three of them, [35], [73], and [83], mentioned that CoE (Center-of-Excellence) is important for the overall procedure of business process automation. Only one referred to the approach for the project [73]. Both [73] and [83] revealed that it was important to begin this work paradigm change with a case pilot or a proof-of-concept (PoC).

On the other hand, the top models are the following five works: [47] and [75] with eleven aspects checked; [39] mentioned nine items; in [38] and [40], eight considerations are checked. Three of them considered using CoE and PoC [39, 47, 75] in the procedure for business process automation. Only one [40] checked the approach for the project (APPR) with Agile. [38], [39], [47], and [75] leave the approach to development out of the process.

However, [40] failed in other steps, more in the post-implementation of automation. [38] studies the overall process improvement procedure but neglects aspects like the involvement of the human resources (roles), methodology for implementation, approach for the project, change management, and plan to upscale. The study of the process of automating is present in all these papers (Yes, in the PROC). [75] also failed in the lack of consideration when choosing the technology or tool for automation. In the model of [47], the authors of this SRL missed the formalization of the stage of monitoring the solution achieved. Integration with other information systems, the assessment of automation, and change management were forgotten by [39].

Relative to the aspects more cited in the description of the models, the authors of this SLR counted the elements presented in each of the 28 studies (Figure 26). PROC defines the top five with 26 references, DEV with 24, DEPL with 18, TEST with 17, and MONI with a count of 16 references. These top five revealed that the authors of the analyzed studies are concerned about general aspects of the procedure: documenting the as-is and to-be scenario (PROC), development of the automation (DEV), test of the development (TEST), the next phase is making the deployment of the process automated (DEPL), and then happens the phase of monitoring the solution (MONI).

However, since 2020, the aspects of roles in the procedure (ROLL), methodology for the procedure (MET), approach for the project (APPR), assessment of automation (ASSE), change management (CHGM), and upscaling the automation (UPSA) started to being noticed in the studies of this area of knowledge.

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TABLE XXVII
COMPARISON OF ASPECTS THAT WERE CONSIDERED IMPORTANT FOR AUTOMATION BETWEEN THE RETRIEVED MODELS

Ref	ROL	MET	APPR	PROC	TECH	INTG	DEV	TEST	ASSE	DEPL	MONI	CHGM	UPSA	Total
[50]				Yes	Yes		Yes			Yes	Yes			5
[27]				Yes	Yes	Yes	Yes	Yes		Yes				6
[6]				Yes	Yes		Yes	Yes	Yes	Yes	Yes			7
[16]	Yes			Yes			Yes	Yes		Yes	Yes			6
[8]	Yes			Yes	Yes									3
[37]				Yes		Yes	Yes			Yes				4
[20]				Yes			Yes			Yes	Yes			4
[72]				Yes			Yes	Yes		Yes				4
[44]		Pilot		Yes	Yes		Yes	Yes	Yes		Yes			7
[35]	CoE			Yes		Yes								3
[40]	***Yes***		***Agile***	***Yes***		***Yes***	***Yes***	***Yes***	***Yes***		***Yes***			***8***
[83]	CoE	Pilot										Yes		3
[48]				Yes			Yes			Yes	Yes			4
[38]				***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***			***8***
[88]				Yes	Yes		Yes		Yes					4
[39]	***CoE***	***PoC***		***Yes***	***Yes***		***Yes***	***Yes***		***Yes***	***Yes***		***Yes***	***9***
[75]	***CoE***	***PoC***		***Yes***		***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***11***
[73]	CoE	PoC	Agile											3
[25]	CoE			Yes			Yes	Yes	Yes	Yes			Yes	7
[31]			Agile	Yes			Yes	Yes		Yes	Yes			6
[61]	CoE	PoC		Yes			Yes	Yes		Yes	Yes			7
[45]		PoC		Yes		Yes	Yes	Yes	Yes	Yes				7
[47]	***CoE***	***PoC***		***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***	***Yes***		***Yes***	***Yes***	***11***
[23]			Lean	Yes	Yes		Yes	Yes	Yes		Yes			7
[53]			Lean	Yes			Yes		Yes		Yes			5
[56]	Yes		Agile	Yes			Yes	Yes		Yes				6
[51]	Yes			Yes	Dev	Yes	Yes			Yes	Yes			7
[58]				Yes	Yes		Yes	Yes	Yes		Yes			6

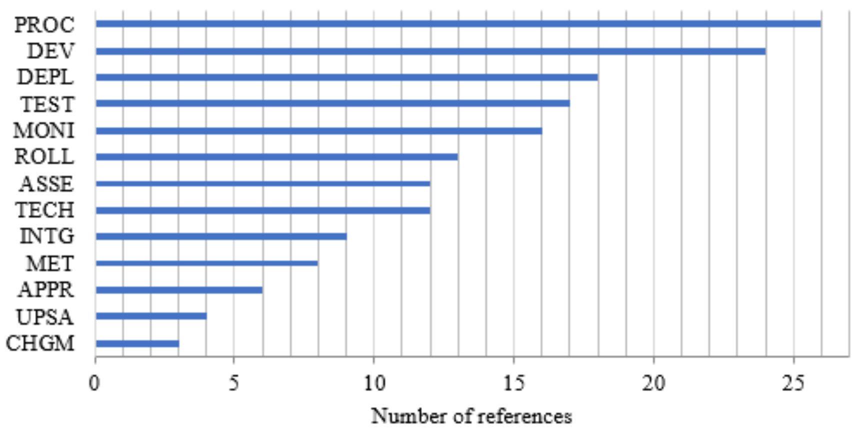

FIGURE 26. Number of papers that checked aspects considered important for automation

IV. CONCLUSIONS

The authors performed an extensive literature review, attempting to understand how the Business Process Automation (BPA) procedure in SMEs was executed. However, this SLR revealed that many studies have been performed in SMEs, but we have still considered the lessons learned. This SLR contributes to creating solid knowledge about the theme and has highlighted some issues that still happen in organizations regarding automation.

This study responded to the five research questions (RQ), and the authors strongly believe this SLR responds to all. (RQ1*) How to typify the processes for automation?* From the 99 analyzed studies, 63 contributed to the answer to this question. The top five process characteristics for automation, extracted from the literature, are occupied by *very repetitive*, *rule-based*, *normalized/structured/ standardized/matured, low complexity (tech without AI)*, and *high number of occurrences*, as represented in Figure 19. Suppose the automation technology has Artificial Intelligence (AI) functionalities; the organization can automate more complex activities (*medium-hight complexity*), semi or *unstructured activities*, or processes that do not yet have *structured data* in their input, like the text of an email or image. This SLR also revealed other characteristics that should be considered in the choice of automation process: *interaction with other*

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systems, *error-prone*, *low level of exception*, and *limited human intervention.* Choosing the perfect candidate for automation can be difficult, and this SLR leaves some references about procedures to perform this task. Ending the answer to this RQ1, this SLR noticed that BPMN 2.0 is still the electable language to design and document the business processes.

(RQ2) *What are the main technologies/tools to support business process automation?* To this answer, the authors of this SLR had contributions from 88 studies. The top three technologies/tools for process automation, extracted from the literature, are occupied with *Robotic Process Automation* (RPA), Intelligent *or Cognitive Process Automation* (IPA or CPA), and *Workflow Manager/Business Process Management System* (WfMS/BPMS), as represented in Figure 20. Some authors mentioned *ERP as an automation facilitator, others mentioned blockchain, and finally, Intelligent Business Process Management Suites* (IBPMS). This work also has an RPA versus I/CPA and an RPA versus BPMS comparison table (Table XIX e XX). These two tables can be a tool for choosing the ideal type of automation technology. Sometimes, the perfect solution can be using various technologies in an integrated way. Technology must be in line with the characteristics of the process to automate.

(RQ3) *What are the main barriers, risks, and limitations in business process automation in SMEs?* For this answer, the authors of this study considered influencing positive and negative factors for BPA; they complement each other. This time, 82 papers were cited in this literature review. Recall that about *40% of RPA projects fail to reach the general goal of decreasing implementation time and implementation costs, among other costs* [23], and *30 to 50% of RPA initiatives fail* [15]. This SLR revealed some factors that can negatively influence BPA (Table XXI): *concerns in cybersecurity and data privacy*; *cost of the solution*; *ethical and moral issues*; *inertia and resistance of employees*; *lack of digitalization in process/organization*; *lack of innovation in the organization*; *lack of knowledge about organizational processes; lack of knowledge about technology/automation*; lack of skilled human resources; lack of solution governance; *lack of strategic alignment (business goals and IT)*; *lack of transparency of the process or change*; *low flexibility/agility in the process*; *not having a methodology for automation*; and *poor management/overcentralized decision-making/rigid hierarchical structure.* The most extracted factors are: *not having a methodology for automation*; *lack of knowledge about technology/automation*; *cost of the solution*; *inertia and resistance of employees*; and *lack of solution governance*, as represented in Figure 21.

On the side of the factors that can positively influence BPA, minimizing the negative ones, Table XXII summarizes them. These factors can be the existence of a *change management plan*; *communication about job disruption*; *the existence of BPM*; *culture/politics/organizational structure that supports changes*; *knowledge about digital technologies*; *low cost of automation solution*; *low time of implementation*; *orchestration of various kinds of BPA approaches*; *plans of communication*; *process management*; and *staff training*. In this case, the top five are occupied with the positive factors: the existence of a *plan of communication*, providing *staff straining* to deal with all the changes, the existence of a *plan for change management*, the existence of a *culture/politics/organizational structure that supports changes*, and *knowledge about digital technologies*. This kind of procedure changes the work paradigm in organizations and requires adaptations in processes and technological infrastructure, so it is extremely important to change management and process management so that the risk of automation failure decreases.

(RQ4) How to evaluate the success of business process automation in SMEs? For the clarification of this theme, 76 studies contributed to this SLR. Process improvement can be measured using Key Performance Indicators (KPIs) or Process Performance M*easures* (PPM). These can be different from one organization to another. This SLR risks affirming that it is impossible to define a set of KPIs suitable for all types of business areas. However, these indicators can be retrieved from BPA's benefits to organizations. So, this SLR resumes all the extracted benefits from the literature in Table XXIII. The most cited benefits are *reducing process time*, *increased effectiveness*, *reduced cost*, *increased accuracy*, and *process improvement*. These top five can be transformed into KPIs in most general organizations. The SLR also discovered the benefits of *employee engagement or satisfaction*, *improving customer satisfaction*, *increased availability*, *mechanisms for auditing*, and *resources for difficult/intelligent tasks*. A large set of KPIs and a performance measurement system can be used to evaluate the improvements created by BPA, so the choice of KPIs should reflect the organization's strategic objectives.

Lastly (RQ5), *What frameworks exist to support process automation for SMEs?* This study compared 28 models (roadmaps, guidelines, methodologies, and frameworks) that can support BPA in organizations. This SRL noticed that there is a possible gap in studies about the application of models to guide the overall procedure of automation of business process automation, specifically in SMEs (only 6 of the 28). This comparison was based on the items: roles in procedure; methodology for the procedure; the approach for the project; study process to automate; technology/tools for automation; integration of the technology information systems; development of the automation; test of automation; assessment of the automation; deployment of the process; monitoring the process; change management and upscaling of the automation. As a result, five works have been selected as the more complete ones: [47] from Herm et al., and [75] from Flechsig et al. with eleven aspects checked; [39] belonging to Noppen et al. mentioned nine items; in [40] from Ludacka et al. and [38] from König et al. there are eight considerations checked. However, [40] failed in other steps, more in the post-implementation of automation. [38] studies the overall procedure of process improvement but neglects aspects like the involvement of the human resources (roles), methodology for implementation, approach for the project, change management, and plan to upscale. The study of the process to automate (PROC) is present in all these papers. [75] also failed in the lack of consideration when choosing the technology or tool for automation. In the model of [47], the authors of this SRL missed the formalization of the stage of monitoring the solution achieved. Integration with other information systems, the assessment of automation, and change management were forgotten by [39].

The top five aspects more cited in the description of the models revealed that the authors of the analysed studies are concerned with general aspects of the procedure: documenting the as-is and to-be scenario (PROC), developing the automation (DEV), testing the development (TEST), the next phase is making the deployment of the process automated (DEPL), and then happens the phase of monitoring the solution (MONI). However, since 2020, the aspects of roles in the procedure (ROLL), methodology for the procedure (MET), approach for the project (APPR), assessment of automation (ASSE), change management (CHGM), and upscaling the automation (UPSA) started to being noticed in the studies of this area of knowledge. So, in conclusion, and with the literature as a base, the authors of this SLR can affirm that all these elements must be included in a model to assist organizations in their path to business process automation.

V. FUTURE WORK

The conclusions of this SLR can be absorbed in several ways. Firstly, they will be used immediately by the authors in the continuation of the study to answer the question, "*What methodological support can be given to SMEs for successfully adopting process automation tools?*" by developing methodological support that allows SMEs to successfully carry out a BPA procedure. This development will be carried out following the Design Science Research (DSR) methodology by Peffers et al. [118], also contributing to an increase in the empirical evidence of these artefacts. In our humble opinion, no study still documents all the intervenients (roles), inputs, outputs, methodologies, approaches, and phases or steps for the automation of business processes to be a success, as can be observed in the answer of the RQ5.

Secondly, the conclusions of this SLR may lead to further research into, for example, (1) the role of Human Resources in the success of BPA procedures; (2) the impact of the level of digital literacy of SMEs on the BPA procedure; (3) the impact of the level of digital transformation of administrative business processes on the BPA procedure; (4) the role of ethics and social responsibility in SMEs when they change their working paradigm (manual to automatic).

We believe that this study has helped to highlight some of the gaps that still exist in the area of BPA and that it is a good basis for future researchers to deepen our knowledge of the subject and, at the same time, to help this type of organization (SMEs), which often don't have the technological, human or budgetary resources to make this journey on their own.

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